

FACULTY OF COMPUTER SCIENCES AND IT
Software Engineering Program Software
Analysis and Design



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Table of Contents

1. Executive Summary
2. System Overview
3. Functional Requirements
4. Non-Functional Requirements
5. System Architecture
6. UML and Software Design Diagrams
7. BPMN and Process Modeling
8. User Interface Prototype
9. Database and Data Flow Design
10. Conclusion

1. Executive Summary

The Hidrosanitare Agolli Management System is a software engineering project designed to digitalize and optimize the business operations of Hidrosanitare Agolli. The project includes multiple software analysis and design artifacts such as BPMN diagrams, Use Case diagrams, Sequence diagrams, Activity diagrams, ER diagrams, UI prototypes, and management models. The repository demonstrates a complete software analysis and documentation workflow used during the software engineering lifecycle.

2. System Overview

The system focuses on inventory management, customer interaction, product handling, sales management, reporting, and administrative control. The repository contains different software engineering models and visual diagrams used to describe business processes, data flows, and system interactions.

3. Functional Requirements

- User authentication and authorization
- Inventory and stock management
- Product and category management

- Sales and invoice processing
- Customer and supplier management
- Administrative reporting
- Workflow and activity management
- Database interaction and management

4. Non-Functional Requirements

- Security and authentication
- Performance and scalability
- Reliability and maintainability
- Usability and accessibility
- Database consistency
- Cross-platform accessibility

5. UML and Software Design Diagrams

Diagram 1: AcceptanceCriteria ReleasePlan

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

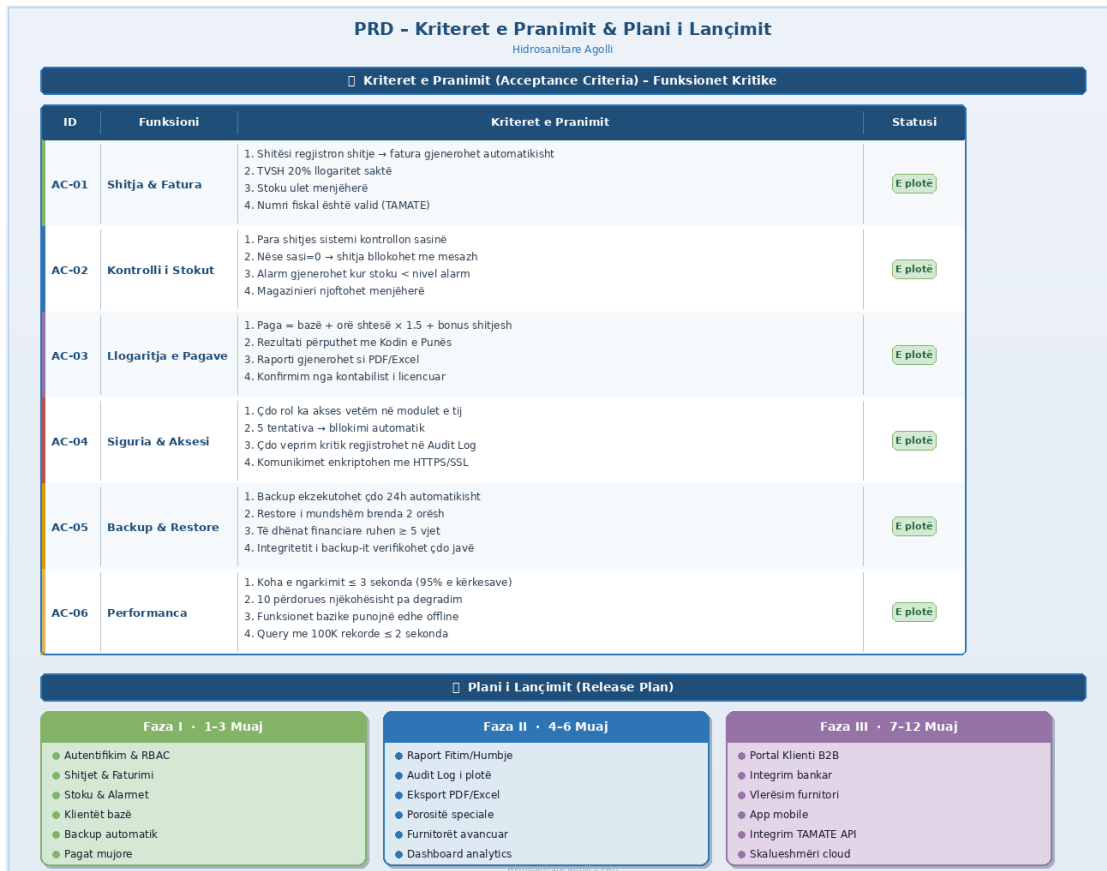


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 2: Assumptions Table

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2.3 Assumptions - Supozimet e Sistemit		
Sistemi i Menaxhimit Hidrosanitare Agolli		
Kategoria	Supozimi	Ndikimi nëse ndryshon
Mjedisi Operacional	Kompania disponon infrastrukturë kompjuterike minimale (LAN, internet)	Mund të kërkojë infrastrukturë shtesë
Mjedisi Operacional	Energjia elektrike dhe interneti disponohen gjatë gjithë orarit të punës	Nevojitet UPS dhe lidhje rezervë
Mjedisi Operacional	Kompania siguron hapësirë serverike (fizike ose cloud) për DB dhe backup	Ndryshohet arkitektura e sistemit
Përdoruesit	Të gjithë përdoruesit janë punonjës ose partnerë të autorizuar nga kompania	Nevojitet sistem vërtetimi më i rreptë
Përdoruesit	Punonjësit marrin trajnim bazik përpara aktivizimit të sistemit	Rritet koha e adaptimit dhe gabimet
Të Dhënat	Çdo produkt ka Product ID unik të paracaktuar nga kompania	Rishikohet skema e identifikimit
Të Dhënat	Valuta kryesore është ALL; TVSH = 20% sipas legjislacionit shqiptar	Përditësohet logjika e llogaritjes
Integrimi	Nuk kërkohet integrim i menjëhershëm me sisteme tatimore ose bankare	Shtohet modul integrimi në fazë të re
Integrimi	Fatura shtypet nëpërmjet printerëve lokalë të lidhur me sistemin	Nevojitet modul printimi cloud
Magazina	Çdo hyrje/dalje mallrash pasqyrohet menjëherë në sistem nga magazinieri	Rrezik mospërputhje stoku fizik/digjital
Magazina	Alarmi stok i ulët aktivizohet automatikisht nën nivelin alarm	Nevojitet rishikim i parametrave alarm

Hidrosanitare Agolli - SRS

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 3: B ClassDiagram.draw.io

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior,

architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

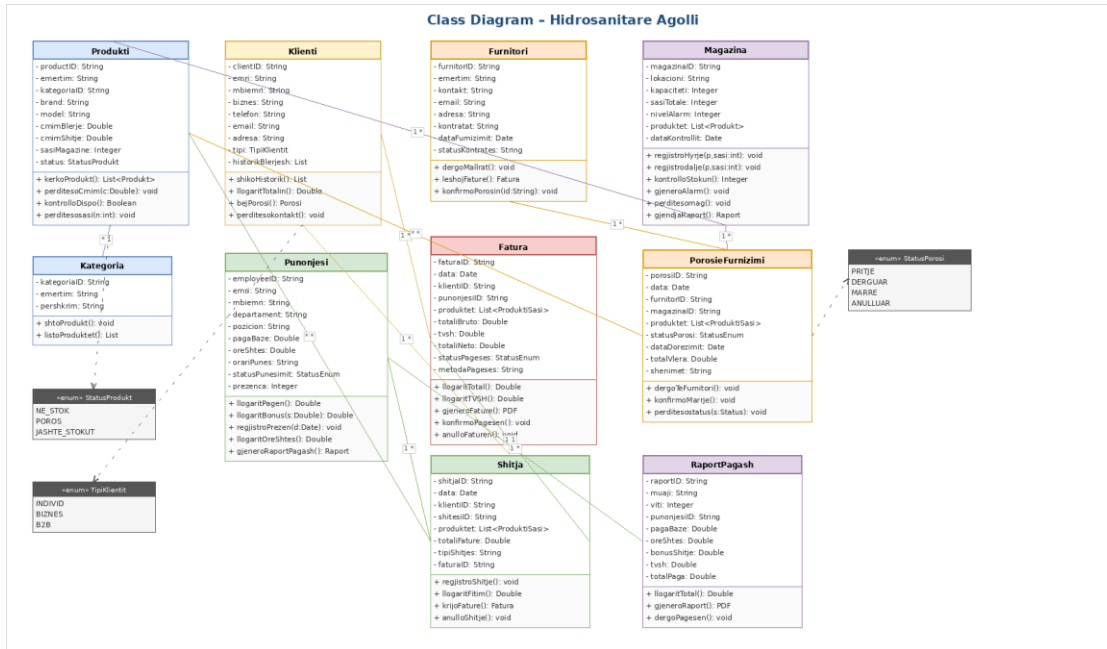


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Diagram 4: Backlog Priorities

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PRD - Product Backlog & Prioritetet					
Hidrosanitare Agolli					
Product Backlog - Kërkesat e Renditura sipas Prioritetit					
ID	Titulli	Përshkrimi (User Story)	Prioriteti	Faza	Statusi
PB-01	Autentifikim & RBAC	Si admin, dua të menaxhoj rolet dhe akseset e çdo përdoruesi.	Kritike	Faza I	✓ Gati
PB-02	Regjistro Shitje	Si shitës, dua të regjistrohem shitjen me produkt, sasi dhe klient.	Kritike	Faza I	✓ Gati
PB-03	Gjenero Faturë TVSH	Si shitës, dua të gjeneroj faturë elektronike me TVSH 20%.	Kritike	Faza I	✓ Gati
PB-04	Kontrollo Stokun	Si sistem, dua të bilokojë shitjen nëse stoku është 0.	Kritike	Faza I	✓ Gati
PB-05	Alarm Stok i Ulët	Si magazinier, dua alarm kur stoku bie nën nivelin minimal.	Kritike	Faza I	✓ Gati
PB-06	Regjistro Hyrje/Dalje	Si magazinier, dua të regjistroj çdo hyrje/dalje mallrash.	Kritike	Faza I	✓ Gati
PB-07	Llogarit Pagat Mijore	Si financiar, dua llogaritje automatike me bonus e orë shtesë.	Kritike	Faza I	✓ Gati
PB-08	Raport Ditor Shitësesh	Si menaxher, dua raport ditor të shitjeve sipas produktit.	E lartë	Faza I	✓ Gati
PB-09	Menaxho Klientët B2B	Si shitës, dua të menaxhoj kontratë e çmime speciale për B2B.	E lartë	Faza I	✓ Gati
PB-10	Porosi Speciale/Import	Si shitës, dua të krijoj porosi kur produkti mungon në stok.	E lartë	Faza I	✓ Gati
PB-11	Raport Fitim/Humbje	Si financiar, dua raport mujor fitim/humbje për vendimmarrje.	E lartë	Faza II	□ Planifikuar
PB-12	Backup Automatik	Si admin, dua backup çdo 24h me mundësi restore brenda 2h.	E lartë	Faza I	✓ Gati
PB-13	Audit Log	Si admin, dua të shoh log-un e çdo veprimi kritik.	Mesatare	Faza II	□ Planifikuar
PB-14	Eksport PDF/Excel	Si financiar, dua të eksporto çdo raport si PDF ose Excel.	Mesatare	Faza II	□ Planifikuar
PB-15	Vlerësim Furnitori	Si magazinier, dua të vlerësoj furnitorin sipas performancës.	E ulët	Faza III	□ E ardhshme
PB-16	Portal Klienti	Si klient B2B, dua të shoh porositë dhe faturat time online.	E ulët	Faza III	□ E ardhshme
PB-17	Integrim Bankar	Si financiar, dua konfirmim automatik pagesash nga banka.	E ulët	Faza III	□ E ardhshme

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 5: C SequenceDiagram Shitja.draw.io

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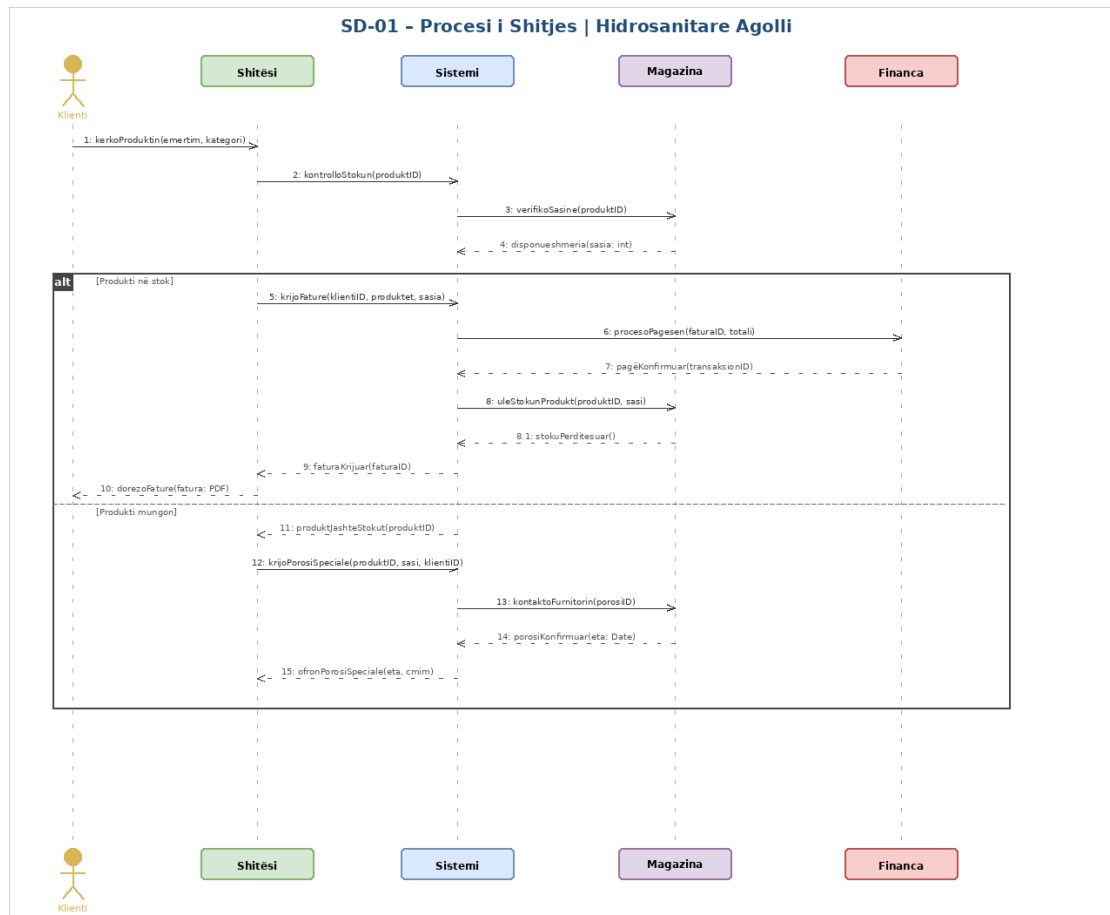


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 6: CollaborationDiagram

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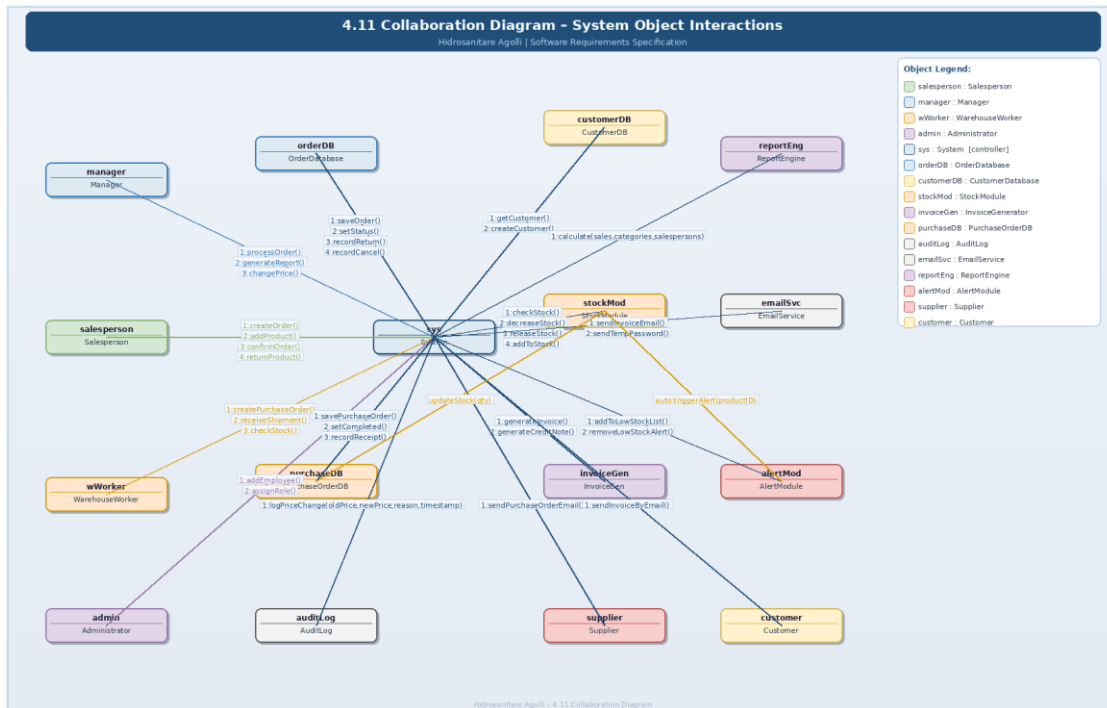


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 7: CompositeStructureDiagram

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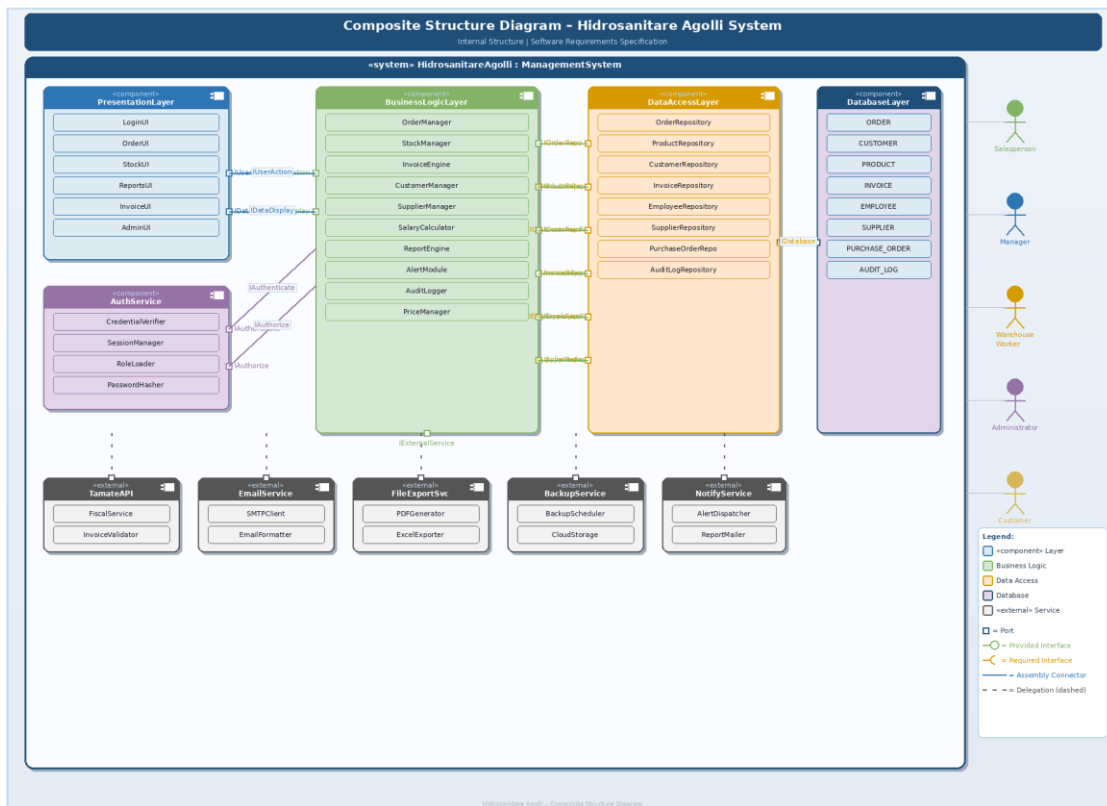


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 8: Constraints Table

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2.4 Constraints - Kufizimet e Sistemit

Sistemi i Menaxhimit | Hidrosanitare Agolli

Kategoria	Kufizimi Kryesor	Impakti
Ligjor	Fiskalizim & TVSH 20% sipas ligjit shqiptar (TATIME)	I lartë
Ligjor	Ruajtja e të dhënave financiare për minimum 5 vjet (Kodi Tatimor)	I lartë
Ligjor	Mbrojtja e të dhënave personale - Ligji Nr. 9887/2008	I lartë
Ligjor	Pagat llogariten sipas Kodit të Punës (orë shtesë, sigurime shoqërore)	I lartë
Teknik	Funksionim offline për shitjet dhe stokun në kushte rrjeti të dobët	Mesatar
Teknik	Mbështetje për 10 përdorues njëkohësisht pa degradim performance	Mesatar
Teknik	Backup automatik çdo 24 orë - ruajtje lokale dhe/ose cloud	Mesatar
Teknik	Enkriptim HTTPS / SSL për të gjitha komunikimet e sistemit	I lartë
Teknik	Koha e ngarkimit të moduleve max 3 sekonda në kushte normale	I ulët
Organizacional	RBAC i rreptë - asnjë përdorues nuk akseson funksione jashtë rolit	I lartë
Organizacional	Audit Log për çdo veprim kritik (fshirje, ndryshim çmimi, anulim fature)	Mesatar
Organizacional	Dokumentim i plotë i sistemit në gjuhën shqipe	I ulët
Organizacional	Implementim pa ndërprerje të plotë të operacioneve - migrim gradual	Mesatar
Financiar	Teknologji open-source ose me licenca ekzistuese për kosto minimale	Mesatar
Kohor	Faza I (shitje, stok, faturim) operacionale brenda afatit të dakordësuar	I lartë
Kohor	Trajnim punonjësish brenda 2 javësh nga aktivizimi i sistemit	I ulët

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 9: DFDdiagram.drawio

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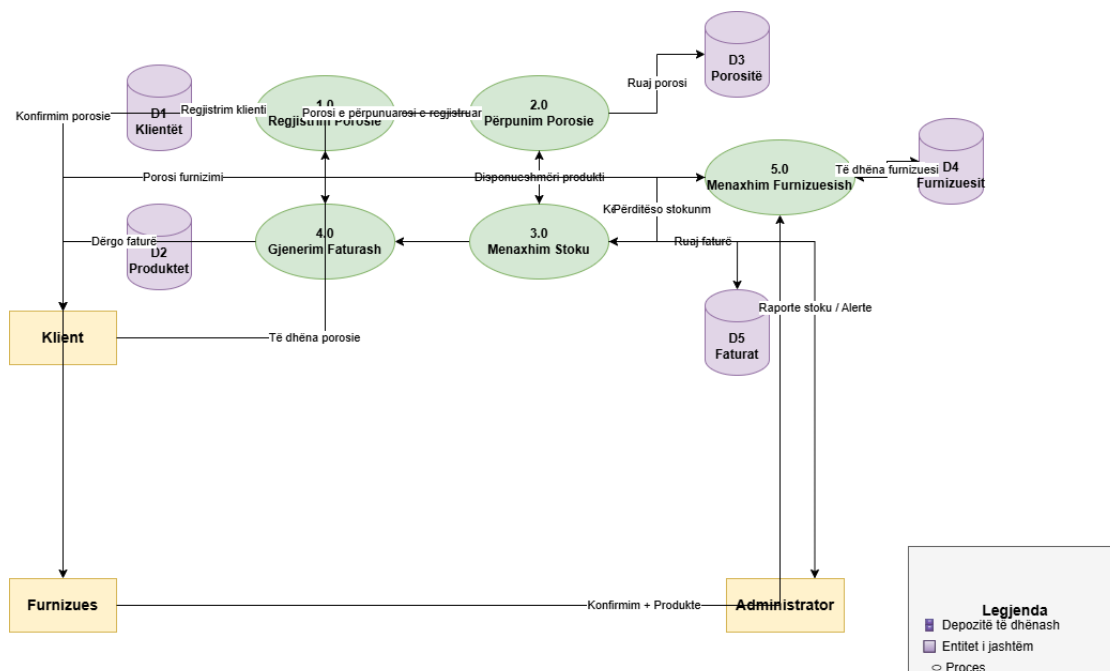


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 10: D SequenceDiagram Pagat.draw.io

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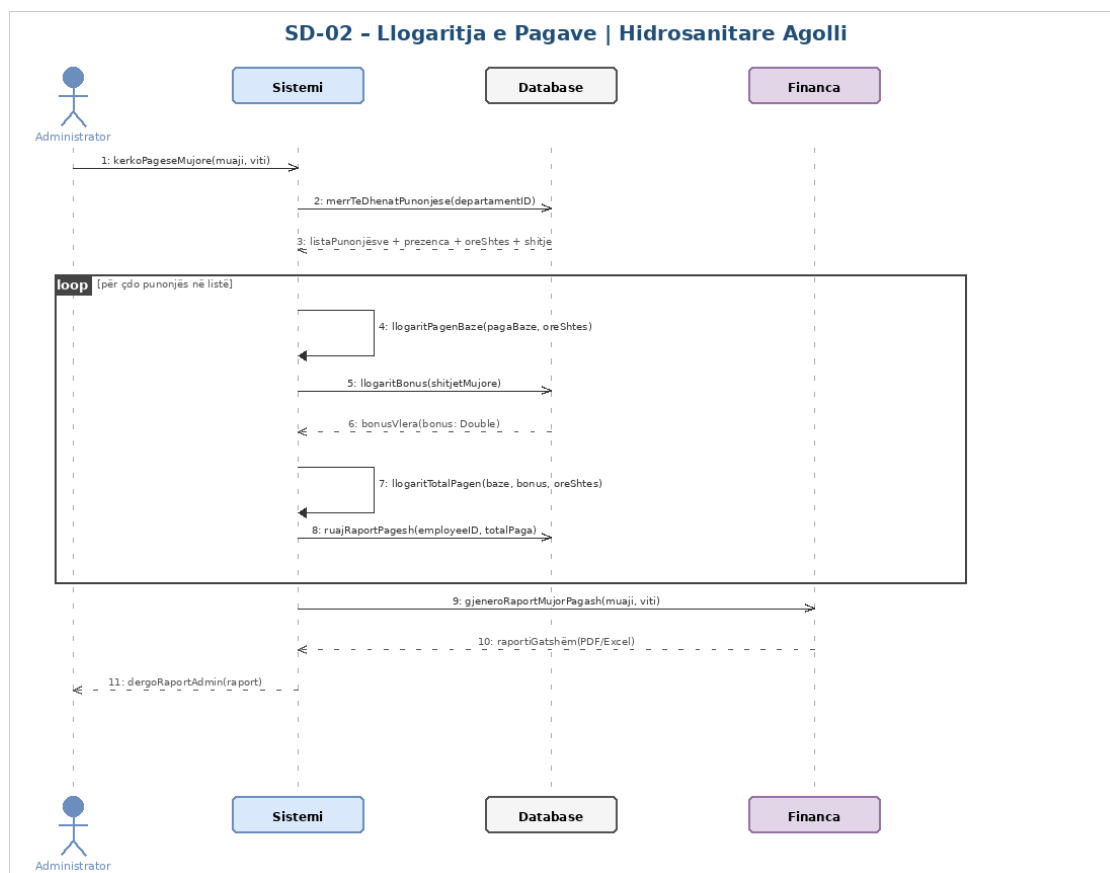


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 11: Dependencies Diagram

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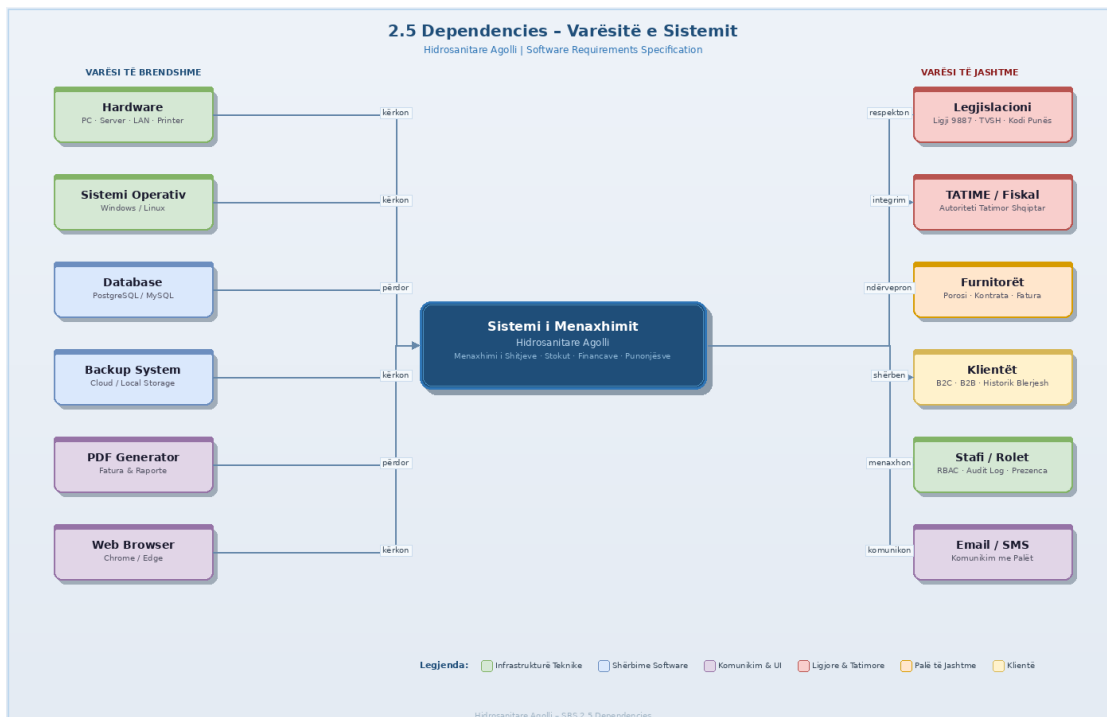


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 12: Deployment.drawio

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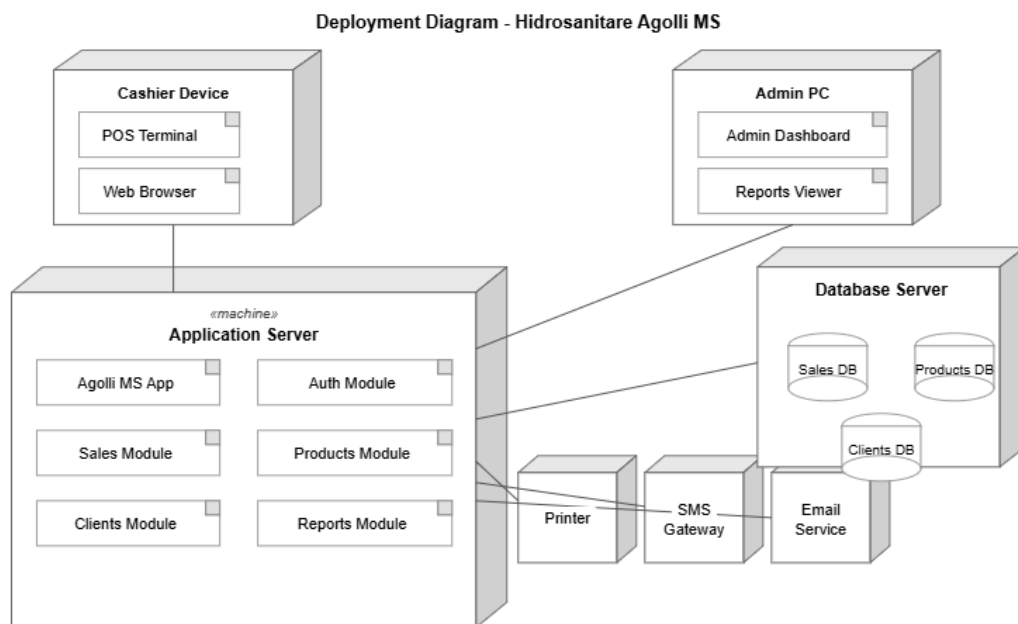


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Diagram 13: Domain Requirements

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3.4 Domain Requirements - Kërkesat e Domainit				
Sistemi i Menaxhimit Hidrosanitare Agolli				
Kërkesat e domainit rrjedhin nga natyra e biznesit të Hidrosanitare Agolli dhe konteksti ligjor, financiar e operacional në tregun shqiptar.				
Rregullat e Domainit (Business Rules)				
ID	Fusha	Rregulli i Domainit	Baza Ligjore	Prioriteti
DR-01	Faturimi & TVSH	Çdo transaksion shitjeje shoqërohet me faturë elektronike me TVSH 20% të llogaritur automatikisht.	Ligji Nr.92/2014	Kritike
DR-02	Numërim Fiskal	Çdo faturë merr numër fiskal unik nga TAMATE. Pa numër fiskal valid transaksioni është i paligjshëm.	Fiskalizimi - TAMATE	Kritike
DR-03	Stok Negativ i Ndaluar	Sistemi bllokoi çdo shitje kur sasia e produktit në magazinë është zero ose negative. Mesazh gabimi shfaqet.	Politika Operative	Kritike
DR-04	Alarm Stok Minimal	Kur sasia bie nën nivelin minimal → alarm automatik + kërkesë furnizimi gjenerohet menjëherë.	Politika Operative	Kritike
DR-05	Llogaritja e Pagave	Paga Neto = Bazë + (Orë Shtesë×1.5×Page/Orë) + Bonus – Sig.Shoq(9.5%) – Sig.Shën(1.7%) – TAR	Kodi i Punës - Neni 102	Kritike
DR-06	Ruajtja Financiare 5 Vjet	Faturat, raportet e pagave dhe raportet mujore ruhen të pandryshueshme për minimum 5 vjet.	Kodi Tatimor - Neni 61	Kritike
DR-07	Çmimet e Produkteve	Sistemi ruan çmim blerje + çmim shitje. Fitimi llogaritet automatikisht. Ndryshimi kërkon autorizim.	Politika Financiare	E lartë
DR-08	Kategoritë e Produkteve	Produktet klasifikohen në: Mishelatorë · Mobjile Tualeti · WC/Bidet · Kabina Dushi · Aksesore · Tubacione.	Klasifikimi Tregtar	E lartë
DR-09	Klientët B2B vs B2C	Klientët B2B kanë çmime kontraktuese dhe afate pagese të ndryshme. Dy profile të ndara në sistem.	Politika Tregtare	E lartë
DR-10	Statuset e Porosive	Porosi furnizimi kalon: Në Pritje → E Dërguar → Marrë & Verifikuar → Mbyllur. Vetëm njëdrejtmisht.	Procesi Logjistik	E lartë
DR-11	Mbrojtja e të Dhënave	Të dhënat personale trajtohen sipas Ligjit 9887/2008. Akses sipas RBAC. Çdo akses regjistrohet.	Ligji 9887/2008	Kritike
DR-12	Valuta & Kursi	Transaksionet kryhen në ALL. Importet me furnitorë ndërkombëtarë regjistrohen me kurs konvertimi manual.	Politika Financiare	Mesatare

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 14: E SequenceDiagram Furnizimi.draw.io

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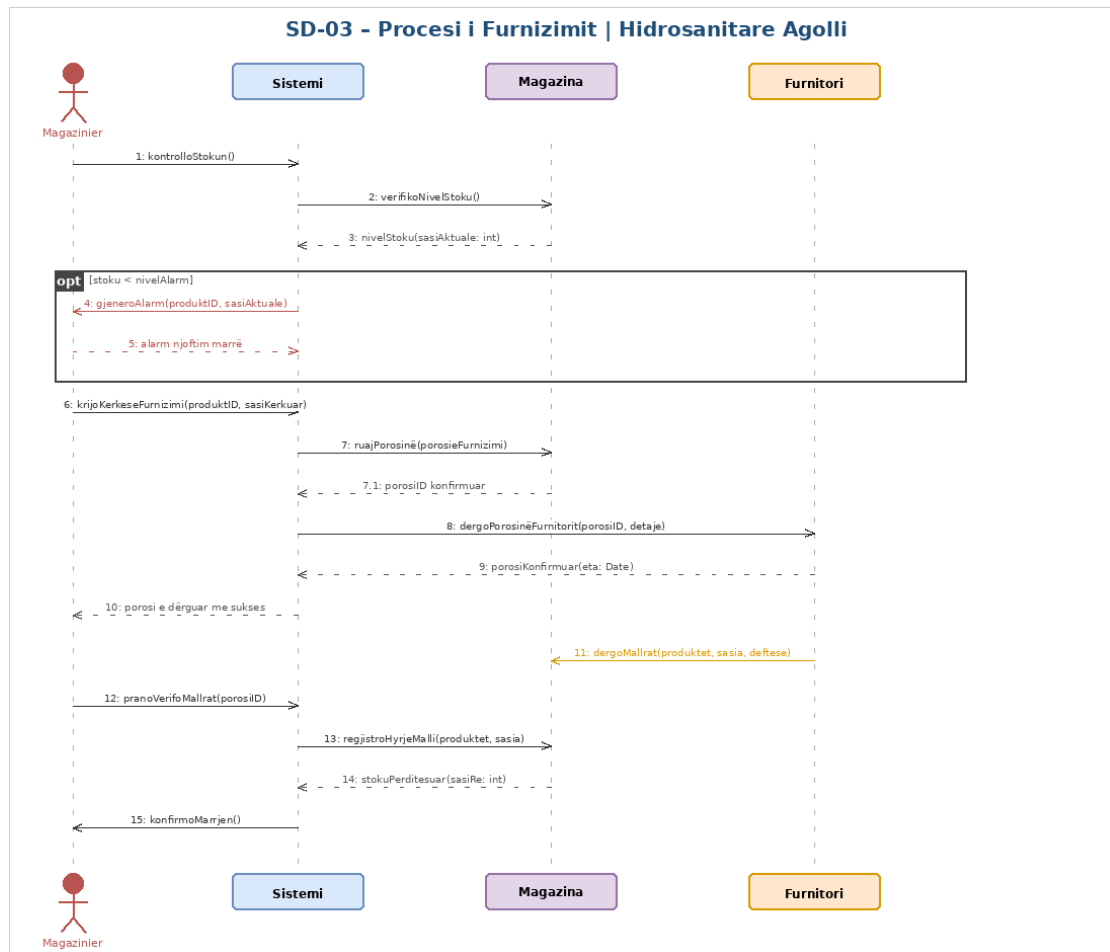


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 15: Features Modules

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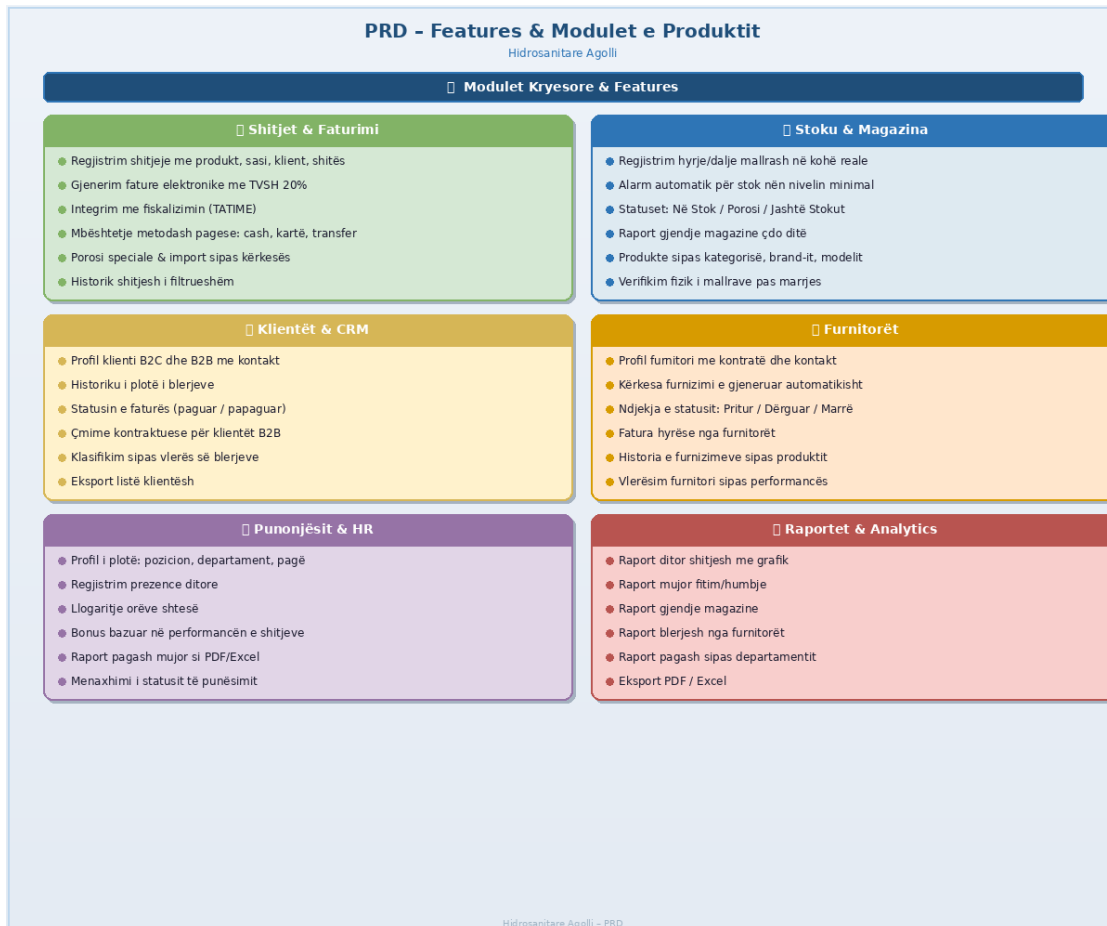


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 16: Goals UserStories

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Diagram 17: Logo hidrosanitary

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Diagram 18: Activity Diagram draw.io

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Klienti kërkon produkt

Shitësi kontrollon stokun

Produkti në stok?

Po

Krijo faturën

Proceso pagesën

Përditëso stokun

Printo faturën

Jo

Krijo porosi speciale

Kontakto furnitorin

Regjistro produktin si në porosi

Produkti dorëzohet klientit

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 19: Sequence Diagram.draw.io

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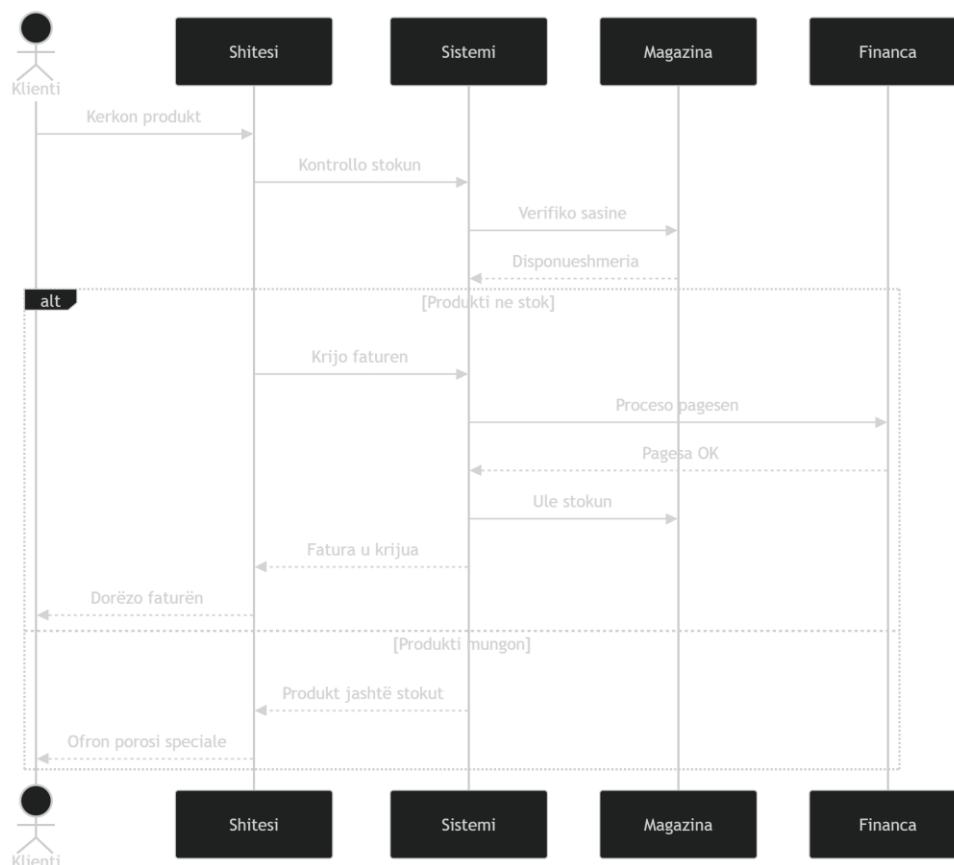


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 20: Use Case Diagram.draw.io

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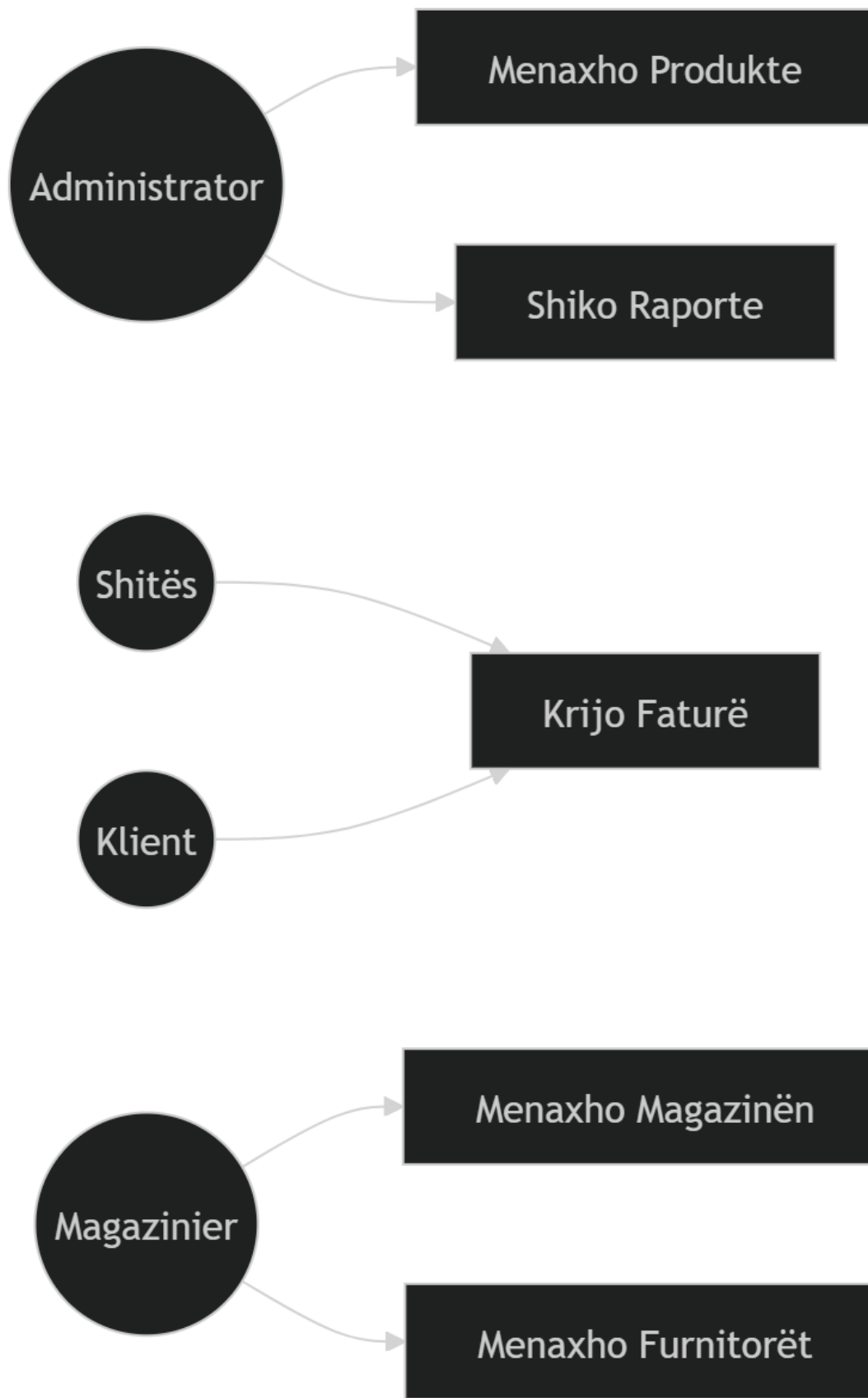


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 21: ad Menaxhimi i Magazinës.draw.io

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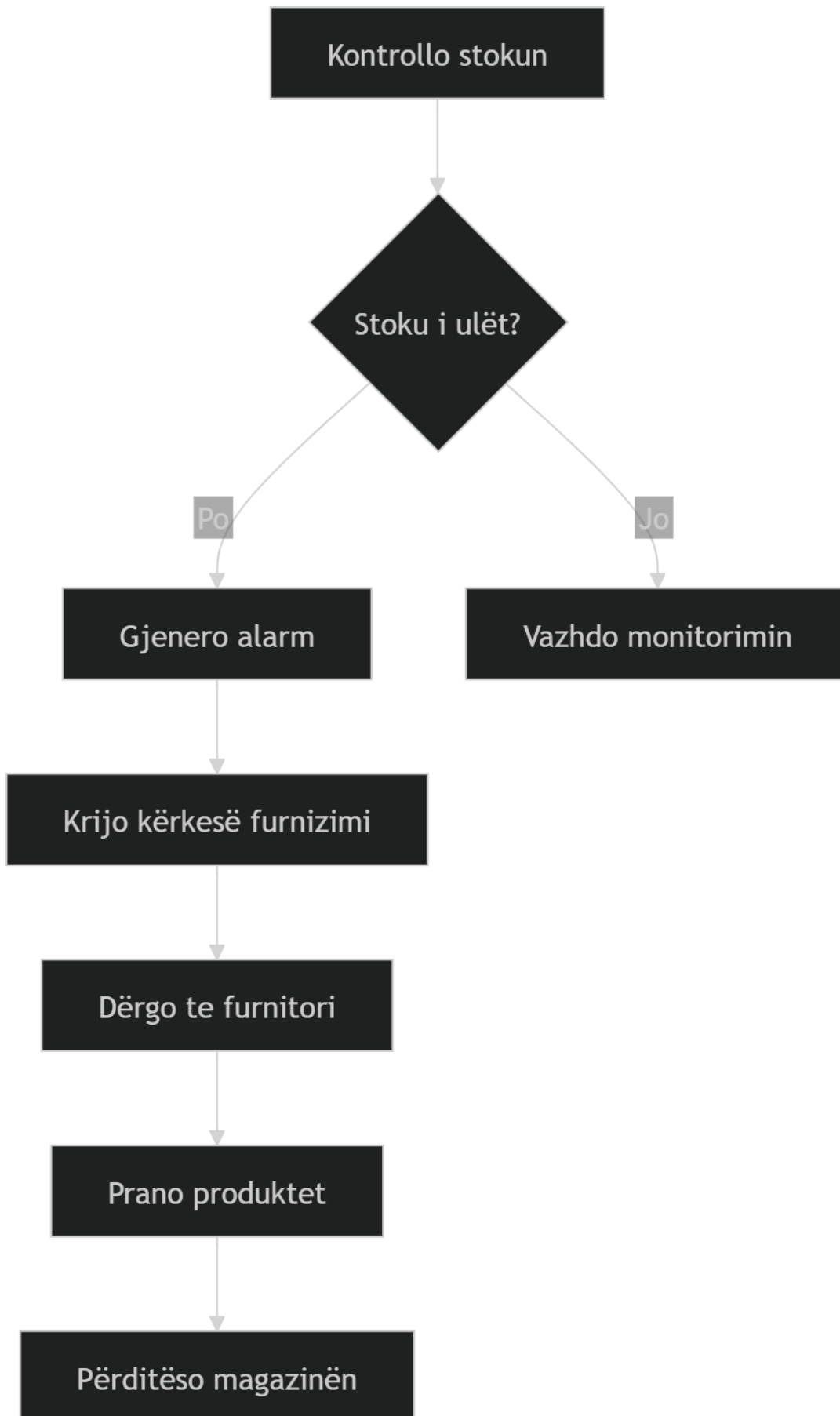


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 22: sd Menaxhimi i Pagave.draw.io

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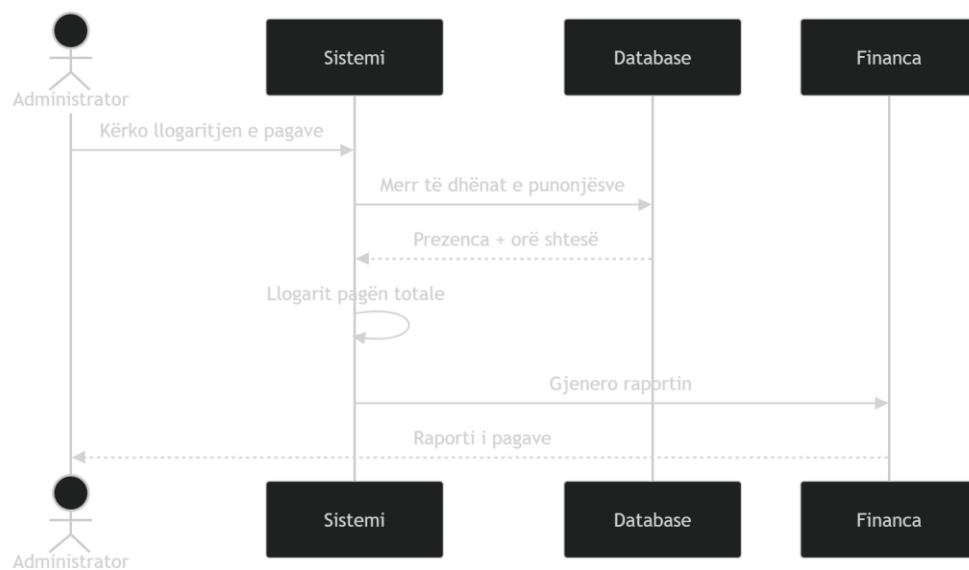


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Diagram 23: Object.drawio

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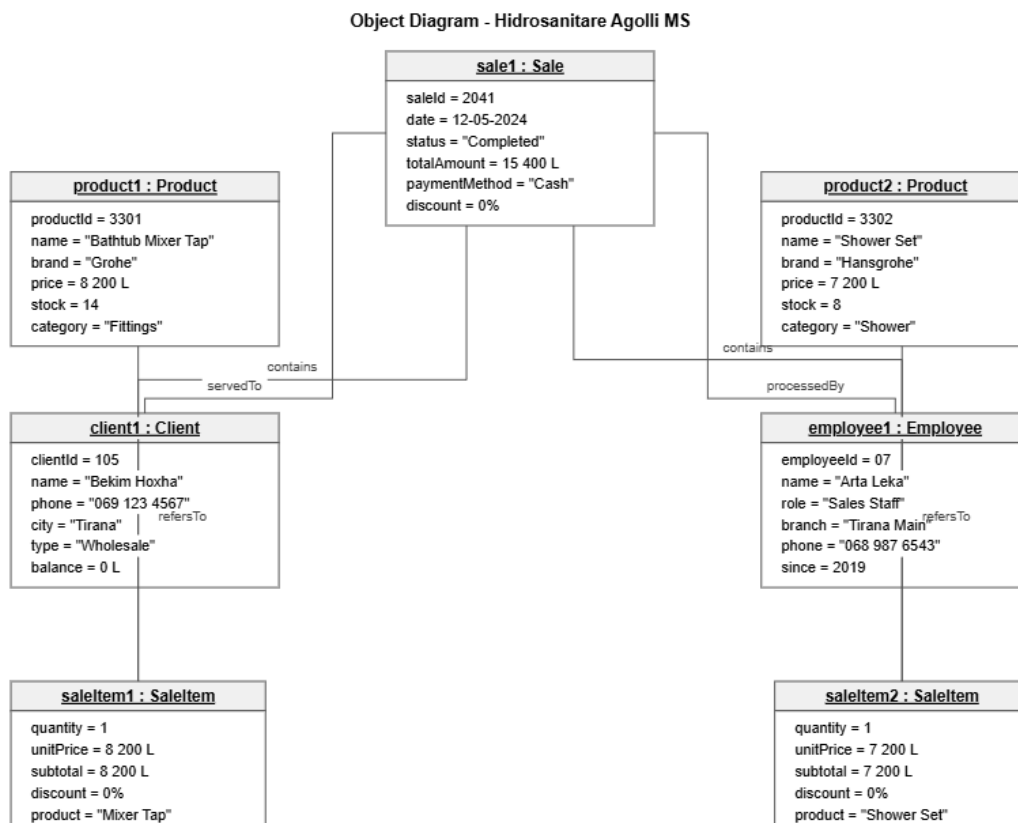


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 24: Onion Diagram

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

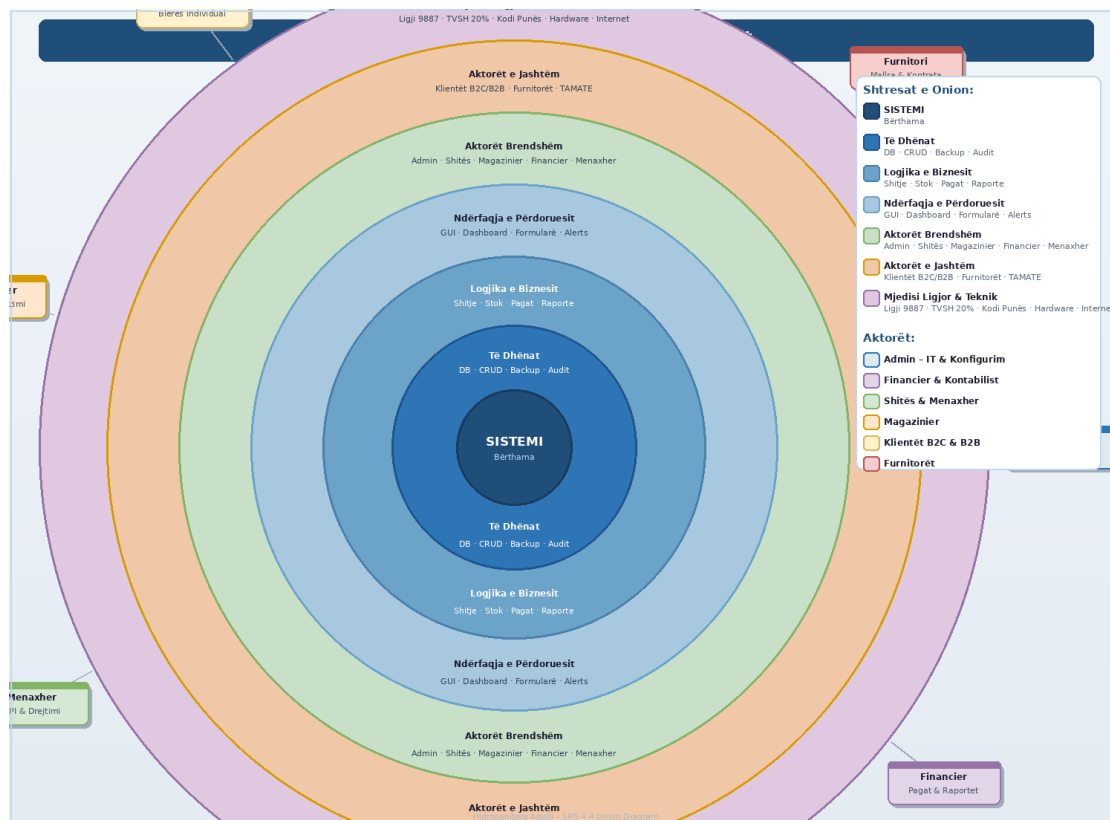


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 25: Onion Layers

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

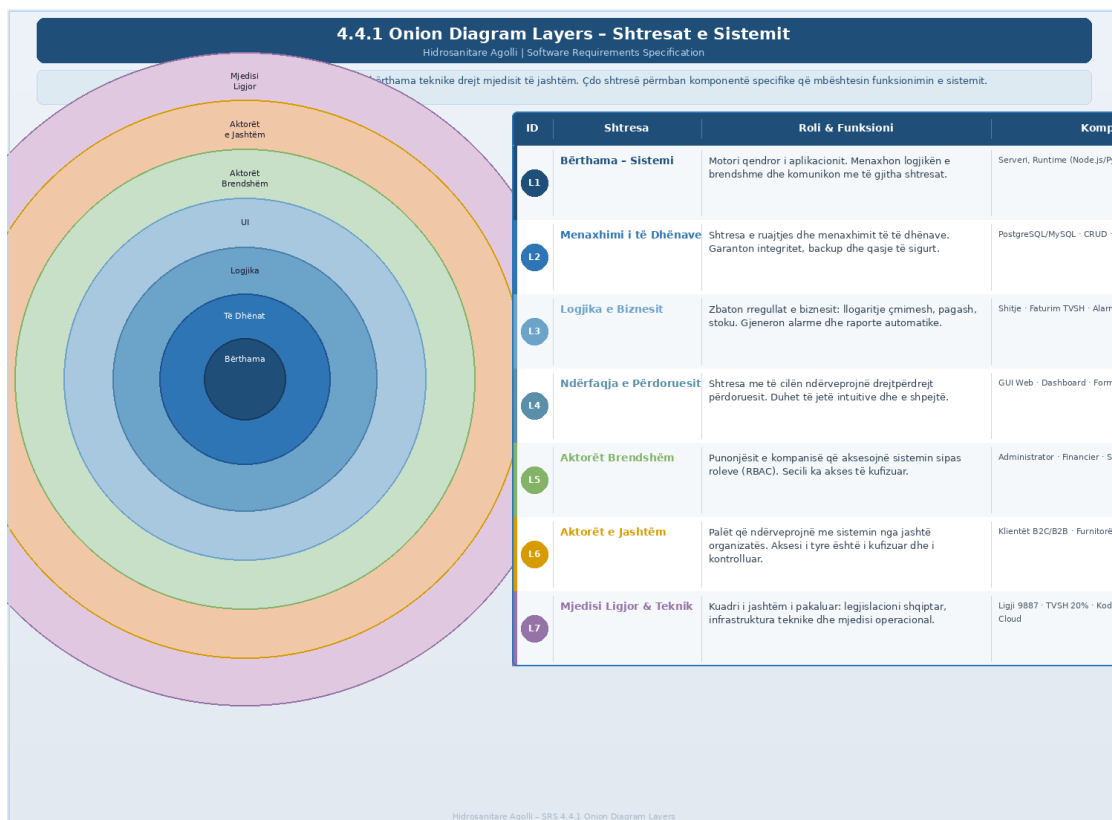


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 26: PRD Overview

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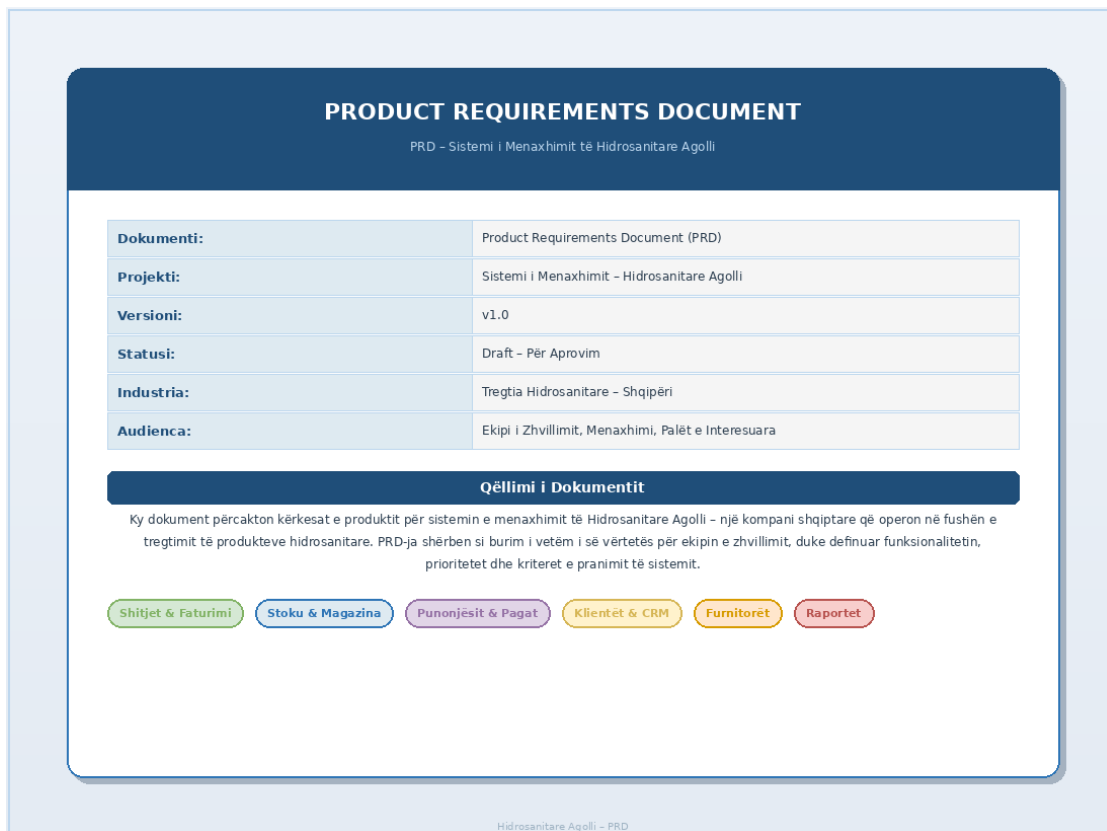


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 27: Markat Tona

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

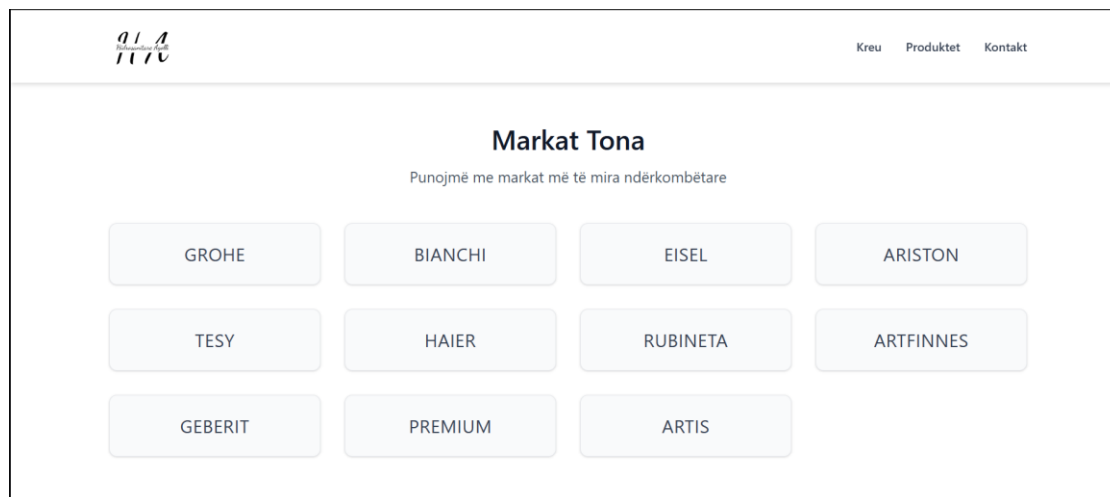


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 28: footer

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or

user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

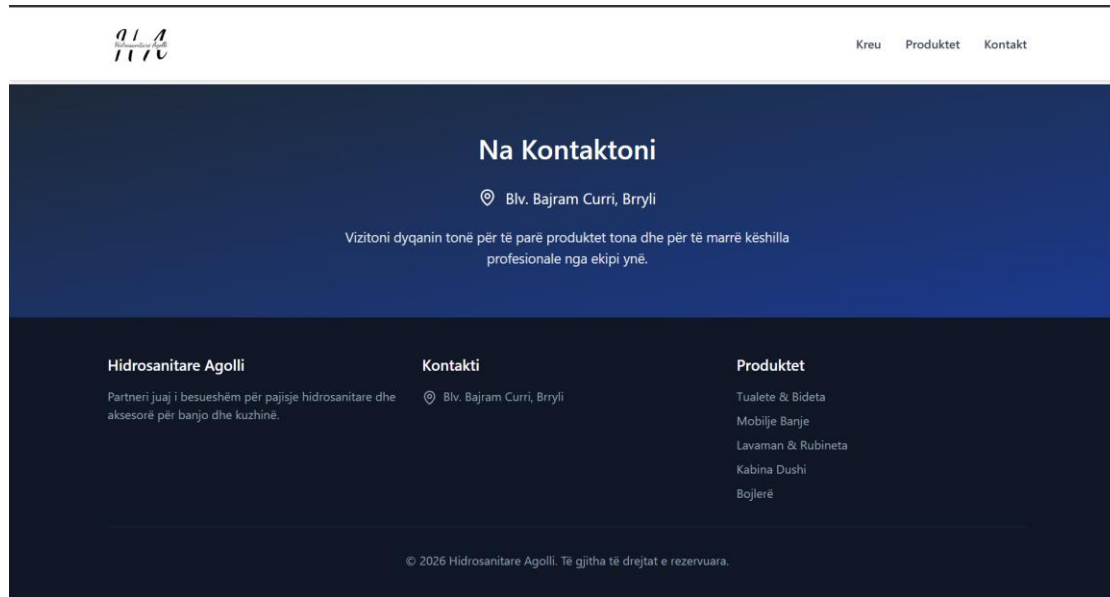


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 29: header

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.



Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 30: kabina dhe xhama

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

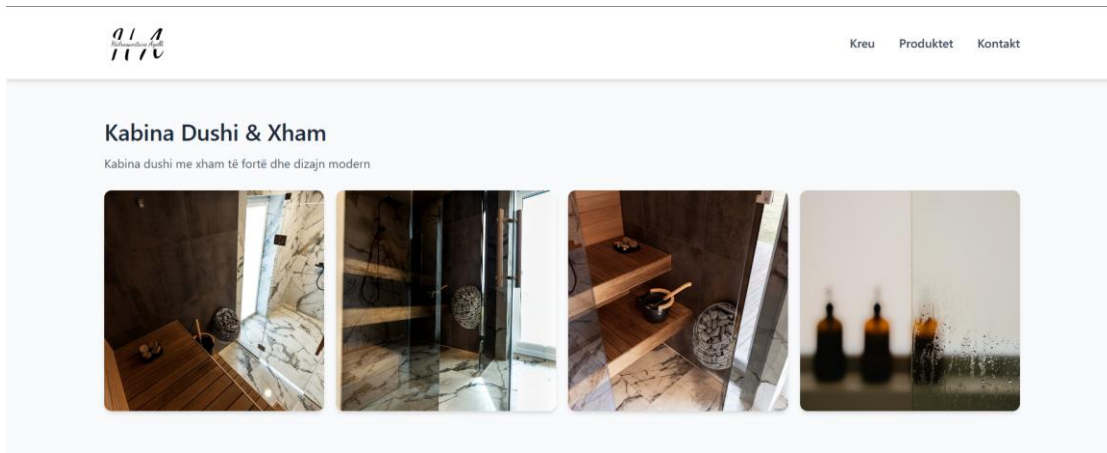


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 31: mobiljetualeti

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

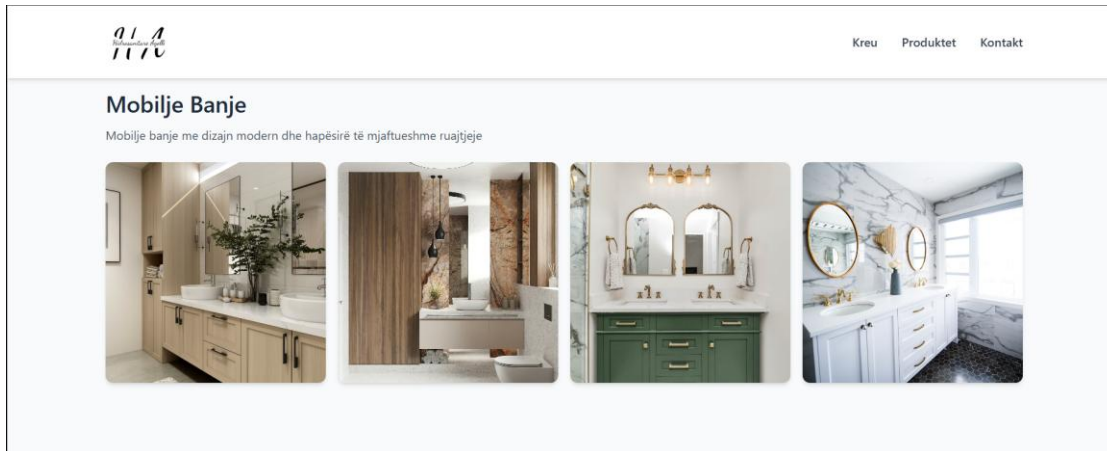


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 32: produkte te tjera

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

Produkte Të Tjera

Bojlerë

Bojlerë elektrikë dhe me gaz me efikasitet të lartë energjetik

Lavaman për Kuzhinë

Lavaman inox dhe granit për kuzhinë

Aksesorë Banje

Mbajtës peshqirësh, çelës, dhe aksesorë të tjerë

Rubinetë Dushi

Rubinetë dhe shifërë dushi me dizajn modern

Valvula Ndërprerëse

Valvula dhe ndërprerës me cilësi të lartë

Lavaman Ngashërimi

Lavaman me dizajn bashkëkohor për banjo

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 33: rubineta

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

Lavaman & Rubineta për Banjo

Lavaman dhe rubineta me dizajn elegant dhe funksionalitet të përsosur

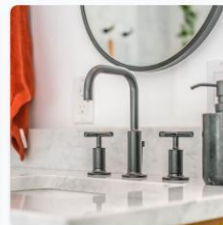
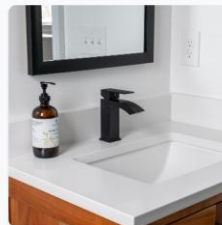
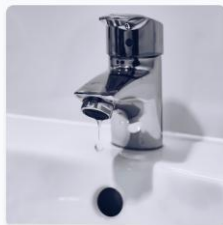


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 34: rubinetakuzhine

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

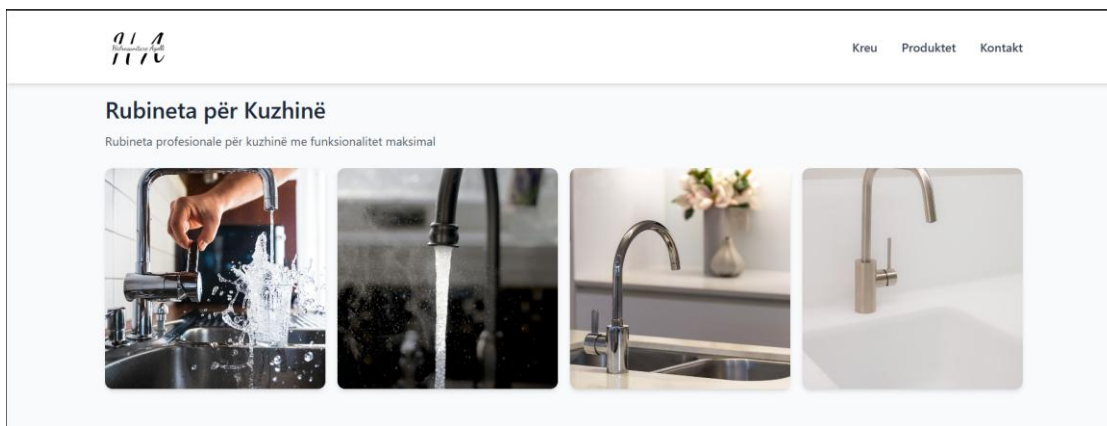


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 35: tualete

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management

System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

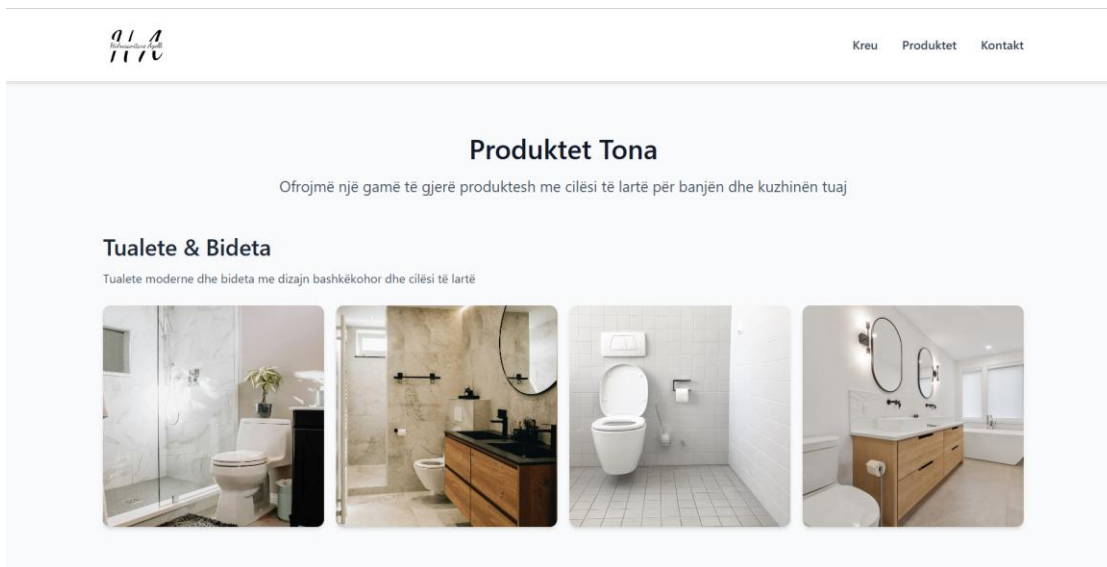


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 36: [UseCaseDiagram.draw.io](https://usecasediagram.draw.io)

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

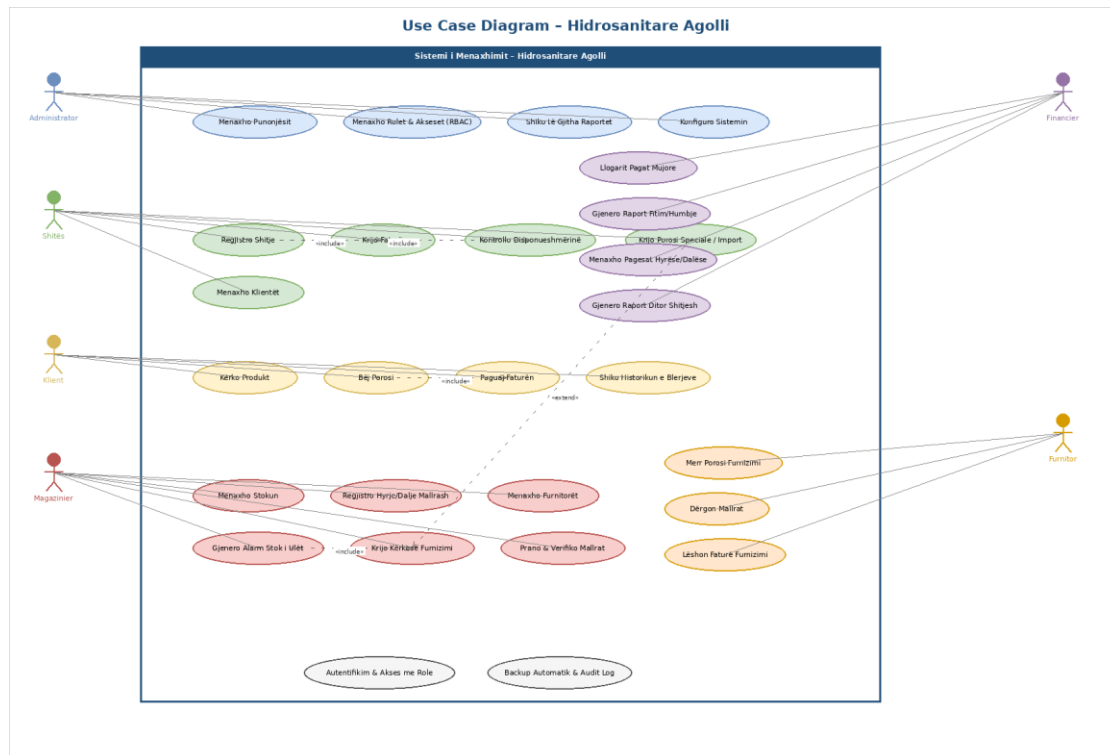


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 37: Use Cases

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

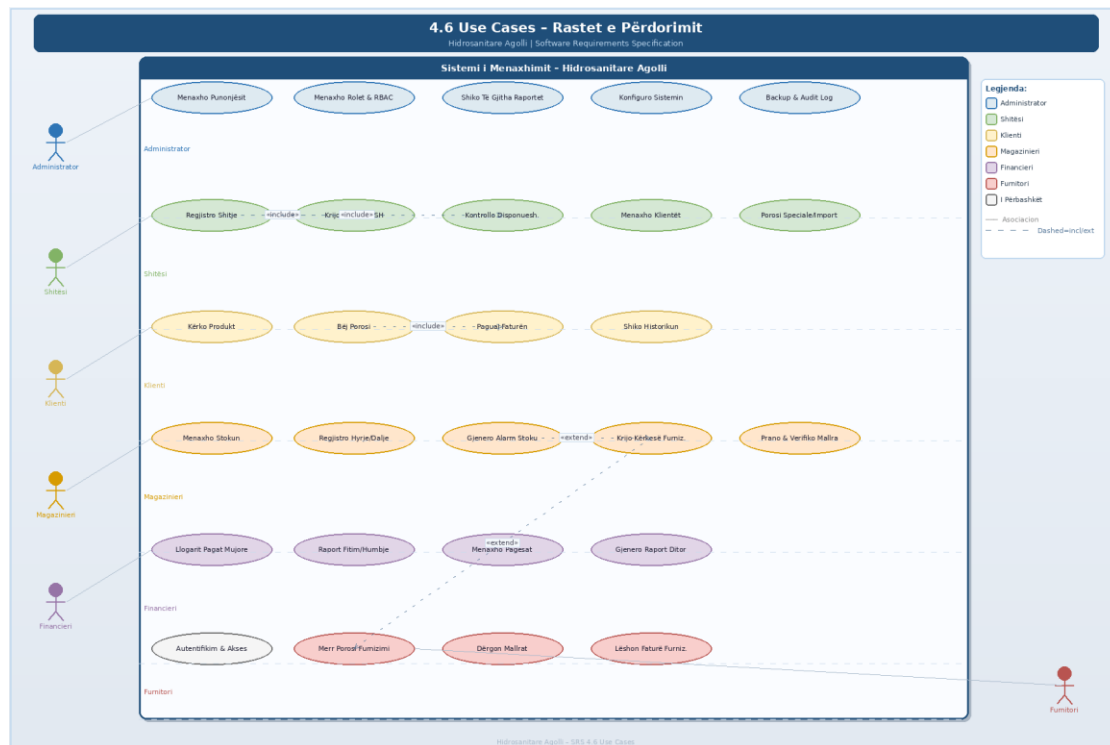


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 38: User Characteristics Table

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

2.2 User Characteristics - Grupet e Përdoruesve			
Sistemi i Menaxhimit Hidrosanitare Agolli			
Grupi i Përdoruesit	Niveli Teknik	Roli & Aksesit	Frekuenca
Administrator	I lartë	Akses i plotë: menaxhon përdoruesit, rolet, konfigurimin dhe backup-in e sistemit	Ditore
Financier / Kontabilist	Mesatar	Gjeneron raporte financiare, menaxhon pagesat hyrëse/dalëse dhe llogarit pagat	Ditore - Mujore
Shitës / Konsulent	Bazik - Mesatar	Regjistrosh shitje, krijon fatura, kontrollon disponueshmërinë dhe menaxhon klientët	Ditore
Magaznier	Bazik	Menaxhon stokun, regjistrosh hyrje/dalje mallrash dhe krijon kërkesa furnizimi	Ditore
Menaxher Mag. / Shitjesh	Mesatar	Monitoron performancën e ekipit, stokun dhe raportet operationale	Ditore - Javore
Klienti (B2C / B2B)	Bazik	Kërkon produkte, bën porosi dhe shikon historikun e blerjeve	Me kërkesë
Furnitori	Bazik	Merr porosi furnizimi, konfirmon dhe lëshon fatura furnizimi	Sipas kontratës

Hidrosanitare Agolli - SR5

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 39: diagram1 magazina.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

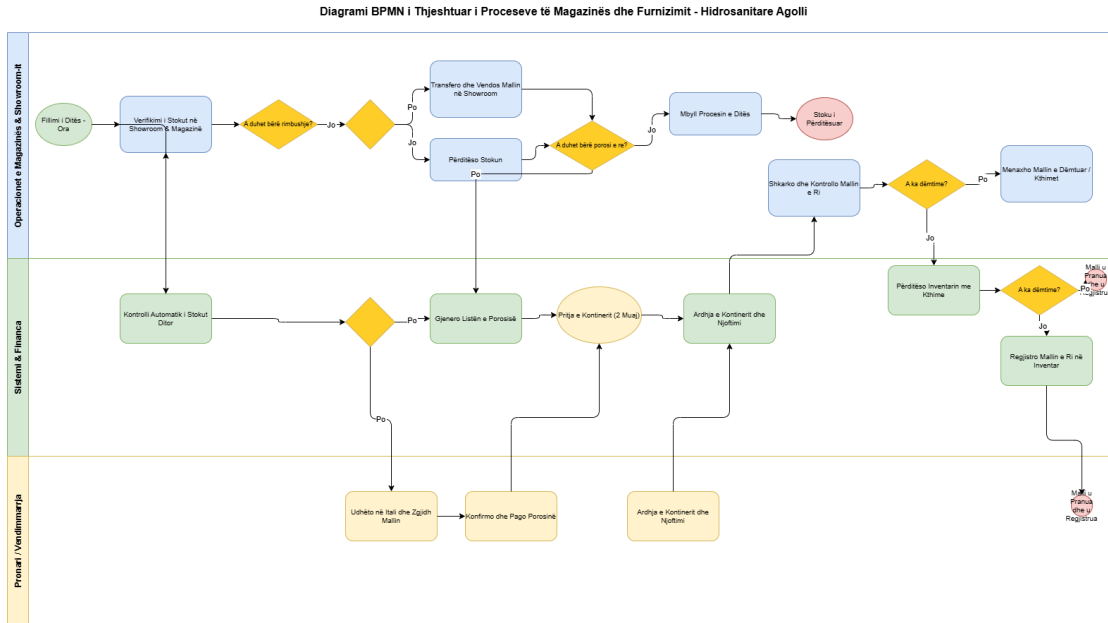


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 40: diagram2 shitja.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

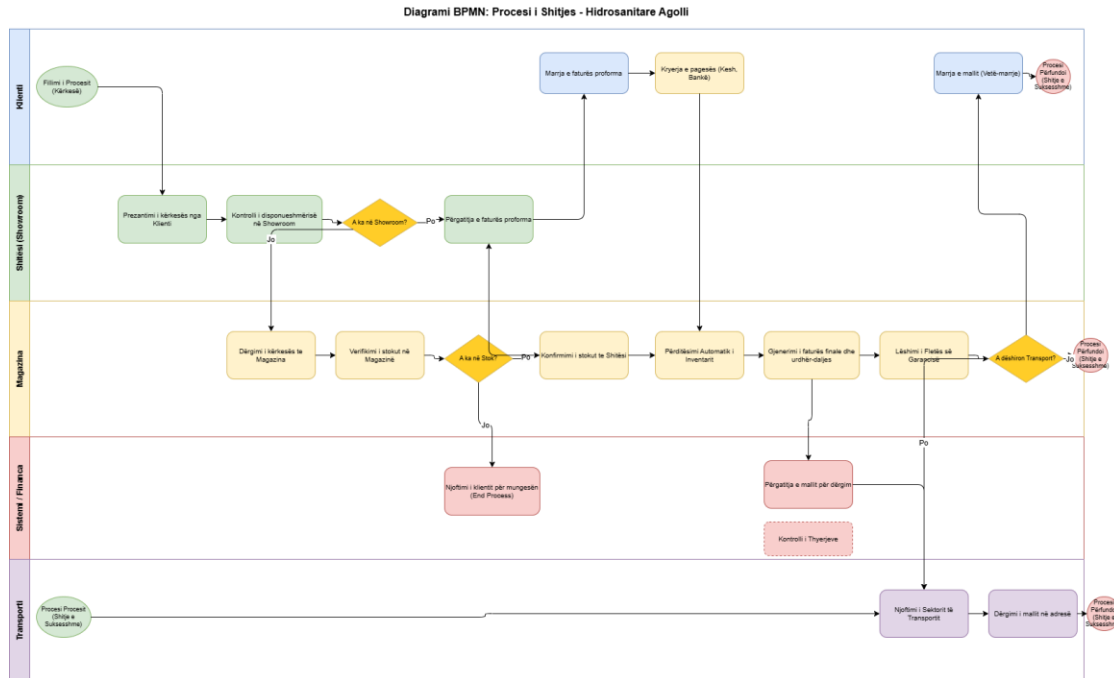


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 41: scenario1.drawio

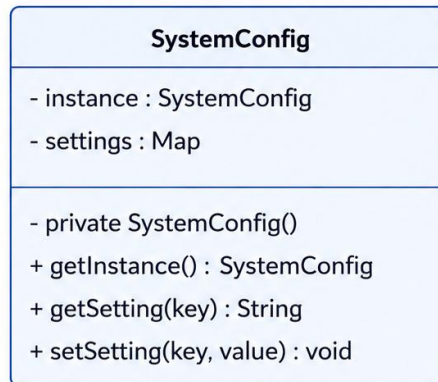
This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

1 Singleton

Qëllimi: Siguron që një klasë ë të ketë vetëm një instancë dhe ofron një pikë globale aksesit për të.

Shembull në sistemin e menaxhimit:

Përdoret për konfigurimet e sistemit.



```
SystemConfig config =  
SystemConfig.getInstance();
```

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 42: scenario10.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

10 Chain of Responsibility

Qëllimi: Shmang lidhjen e dërguesit të një kërkesë me marrësin e saj duke i dhënë mundësi më shumë objekteve ta trajtojnë kërkesën. Kërkesa i kalon zinxhirit derisa të trajtohet nga një prej tyre.

Shembull në sistemin e menaxhimit:

Përdoret për të trajtuar kërkesa për miratim shpenzimesh (expense), ku çdo nivel menaxherial ka një limit miratimi. Nëse një menaxher nuk e miraton (sepse shuma e tejkalon limitin e tij), kërkesa i kalon menaxherit pasues në zinxhir.

Fluksi i ekzekutimit:

- 1 Klienti krijon kërkesën (Expense).
- 2 Dërgon kërkesën te objekti i parë në zinxhir (TeamLeader).
- 3 Secili approver kontrollon nëse mund ta miratojë.
- 4 Nëse jo, e kalon te pasuesi (next.handleRequest(...)).
- 5 Procesi vazhdon derisa të miratohet ose të arrijë në fund të zinxhirit.



Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 43: scenario2.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type.

It contributes to understanding how the platform operates and how different software components communicate.

2 Factory Method

Qëllimi: Përcakton një ndërfaqe për krijimin e objekteve, por lejon nënklasat të vendosin se cilën klasë të instancojnë.

Shembull në sistemin e menaxhimit:

Përdoret për krijimin e raporteve të ndryshme.

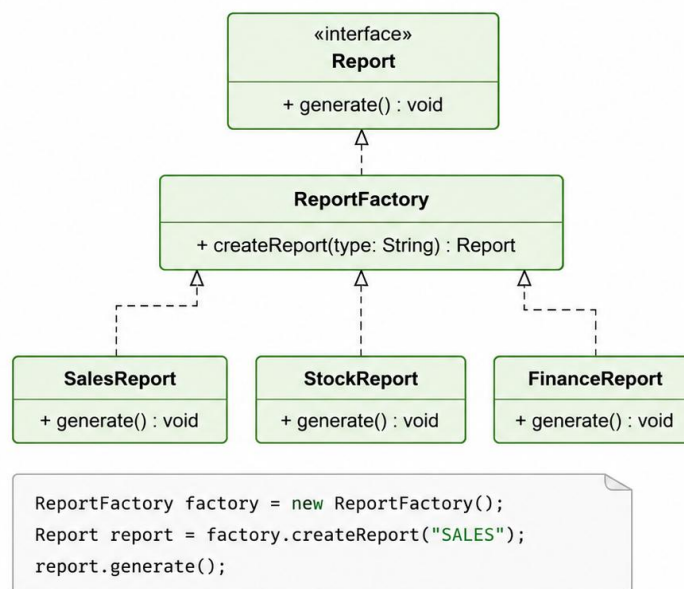


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 44: scenario3.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or

user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

3 Observer

Qëllimi: Përcakton varësi një-me-shumë midis objekteve, në mënyrë që kur një objekt ndryshon gjendjen, të gjithë varësit të njoftohen automatikisht.

Shembull në sistemin e menaxhimit:

Përdoret për njoftimet e stokut të ulët.

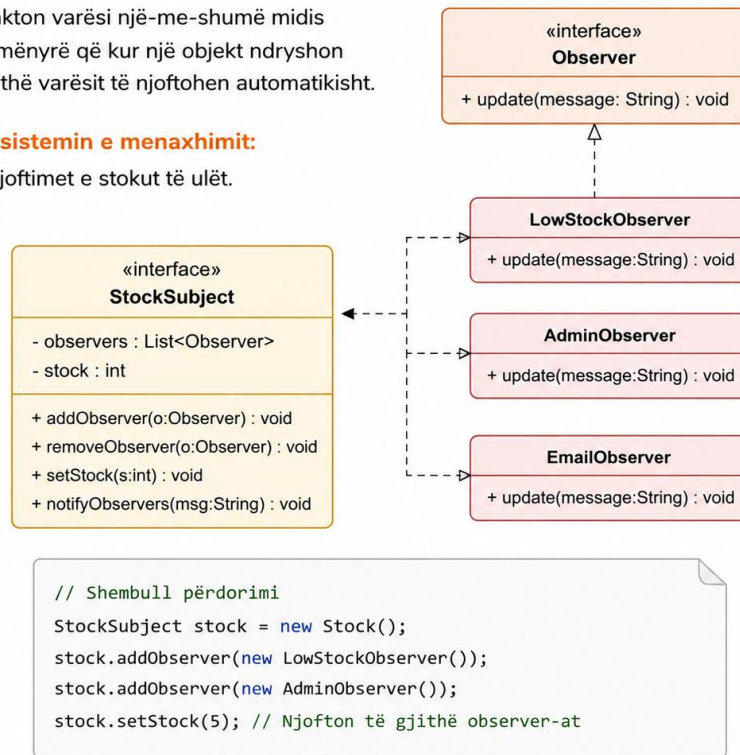


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 45: scenario4.drawio

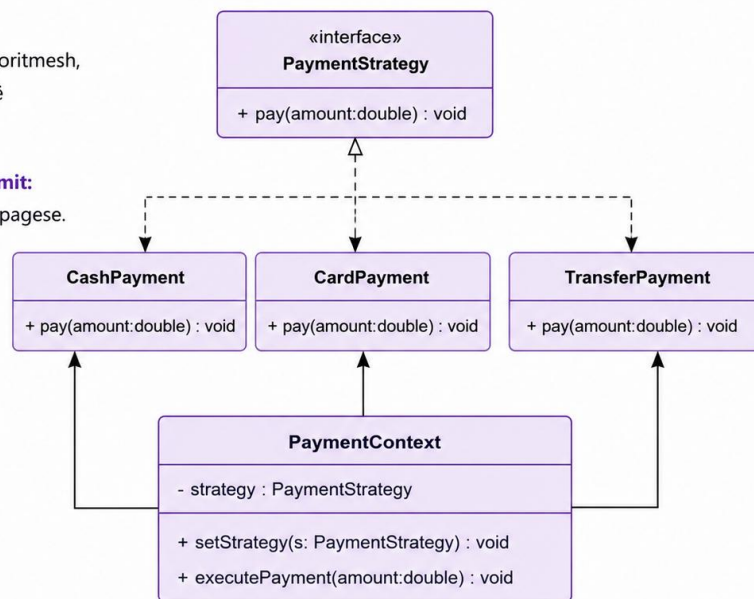
This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior,

architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

4 Strategy

Qëllimi: Përcakton një familje algoritmesh, inkapsulon secilën dhe i bën ato të ndërkëmbyeshme.

Shembull në sistemin e menaxhimit:
Përdoret për metoda të ndryshme pagese.



```
// Përdorimi
PaymentContext context = new PaymentContext();
context.setStrategy(new CardPayment()); // Mund të jetë CashPayment ose TransferPayment
context.executePayment(100.0);
```

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 46: scenario5.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management

System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

5 Decorator

Qëllimi: Shton përgjegjësi shtesë në një objekt dinamikisht, pa ndryshuar strukturën e tij.

Shembull në sistemin e menaxhimit:

Përdoret për të shtuar veçori shtesë në faturë (p.sh. zbritje, TVSH, shënime) pa ndryshuar klasën bazë të faturës.

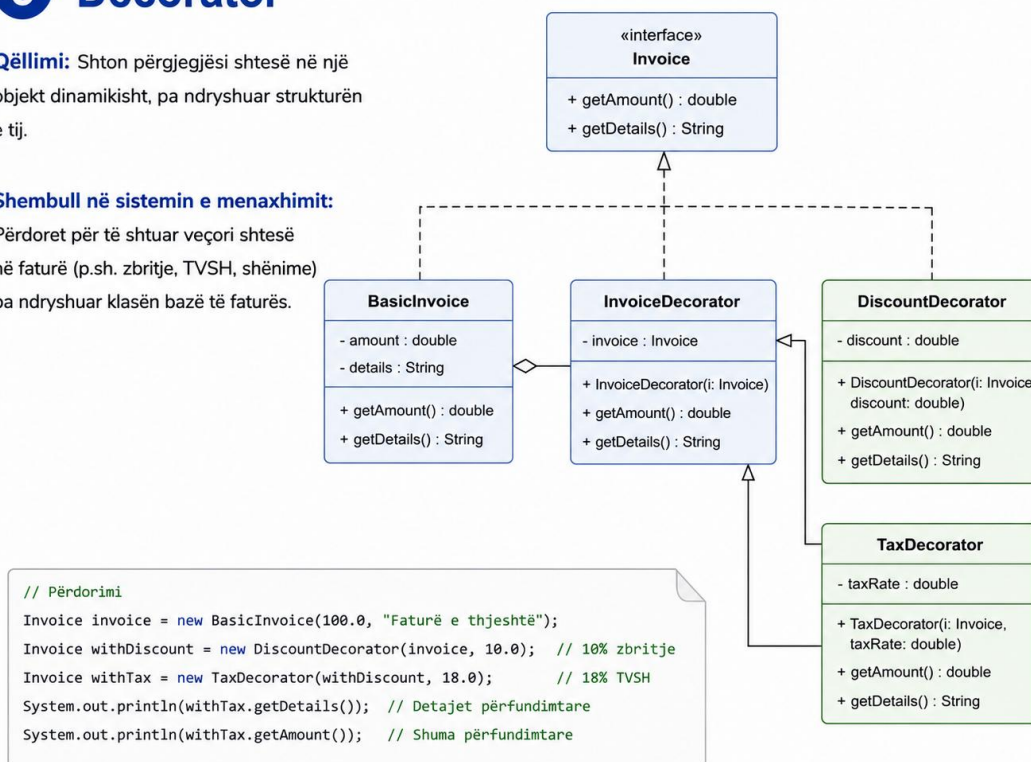


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 47: scenario6.drawio

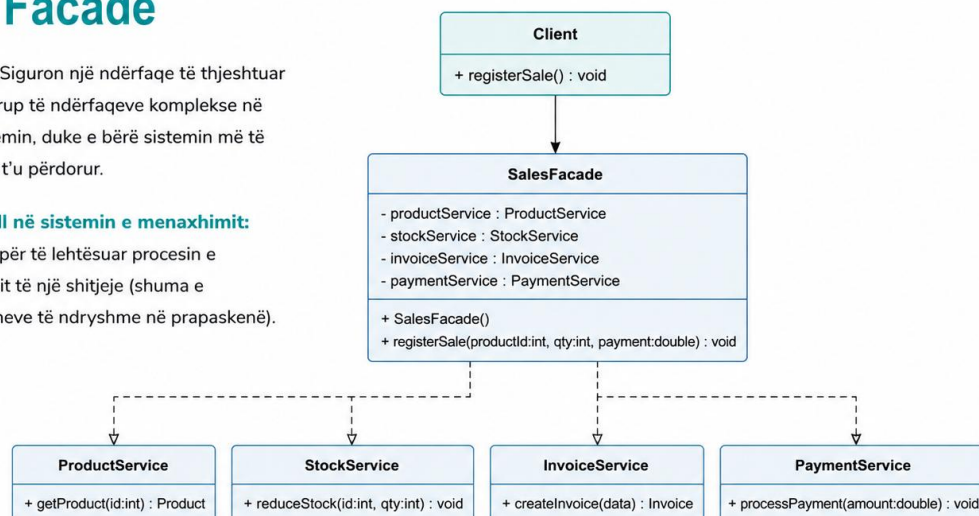
This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

6 Facade

Qëllimi: Siguron një ndërfaqe të thjeshtuar për një grup të ndërfaqeve komplekse në nën-sistemin, duke e bërë sistemin më të lehtë për t'u përdorur.

Shembull në sistemin e menaxhimit:

Përdoret për të lehtësuar procesin e regjistrimit të një shitjeje (shuma e operacioneve të ndryshme në prapaskenë).



```
// Përdorimi i Facade
SalesFacade facade = new SalesFacade();
// Klienti nuk merret me detajet e nën-sistemit
facade.registerSale(101, 2, 1500.0);
// Brenda metodës registerSale bëhen thirrje tek shërbimet e nën-sistemit:
1. Merret produkti
2. Përditësohet stoku
3. Krijohet fatura
4. Kryhet pagesa
```

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 48: scenario7.drawio

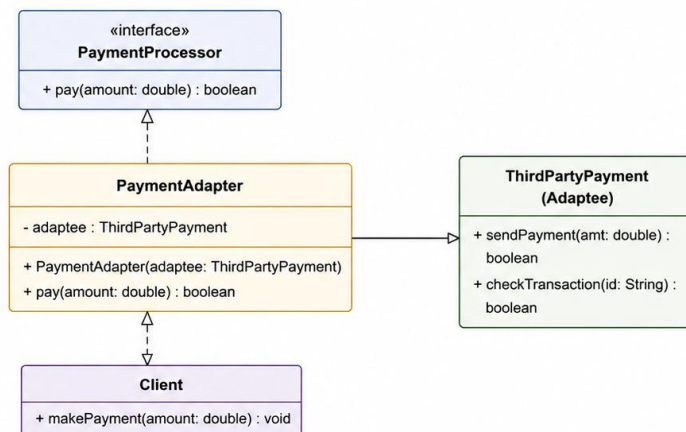
This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

7 Adapter

Qëllimi: Lejon që ndërfaqe të papajtueshme të punojnë së bashku, duke i përshtatur njëra-tjetrën.

Shembull në sistemin e menaxhimit:

Përdoret për të integruar një sistem të pagesave nga një palë e tretë (p.sh. bankë ose gateway) që ka një ndërfaqe të ndryshme, me sistemin tonë ekzistues.



```
// Ndërfaqja që kërkon sistemi ynë
public interface PaymentProcessor {
    boolean pay(double amount);
}

// Klasa ekzistuese e palës së tretë (ndërfaqe e papajtueshme)
class ThirdPartyPayment {
    public boolean sendPayment(double amt) {
        // Logjika e dërgimit të pagesës te palë e tretë
        return true;
    }
}

// Adapteri që e përshtat ndërfaqen
class PaymentAdapter implements PaymentProcessor {
    private ThirdPartyPayment adaptee;

    public PaymentAdapter(ThirdPartyPayment adaptee) {
        this.adaptee = adaptee;
    }

    @Override
    public boolean pay(double amount) {
        // Përdor metodën ekzistuese të palës së tretë
        return adaptee.sendPayment(amount);
    }
}

// Përdorimi
PaymentProcessor processor = new PaymentAdapter(new ThirdPartyPayment());
processor.pay(100.0);
```

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 49: scenario8.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

8 Composite

Qëllimi: Përbën objekte në strukturë në formë peme dhe lejon klientin të trajtojë njësoj objektet individuale dhe grupet e objekteve.

Shembull në sistemin e menaxhimit:

Përdoret për të përfaqësuar strukturën hierarkike të një organizate (p.sh. kompani me departamente dhe punonjës).

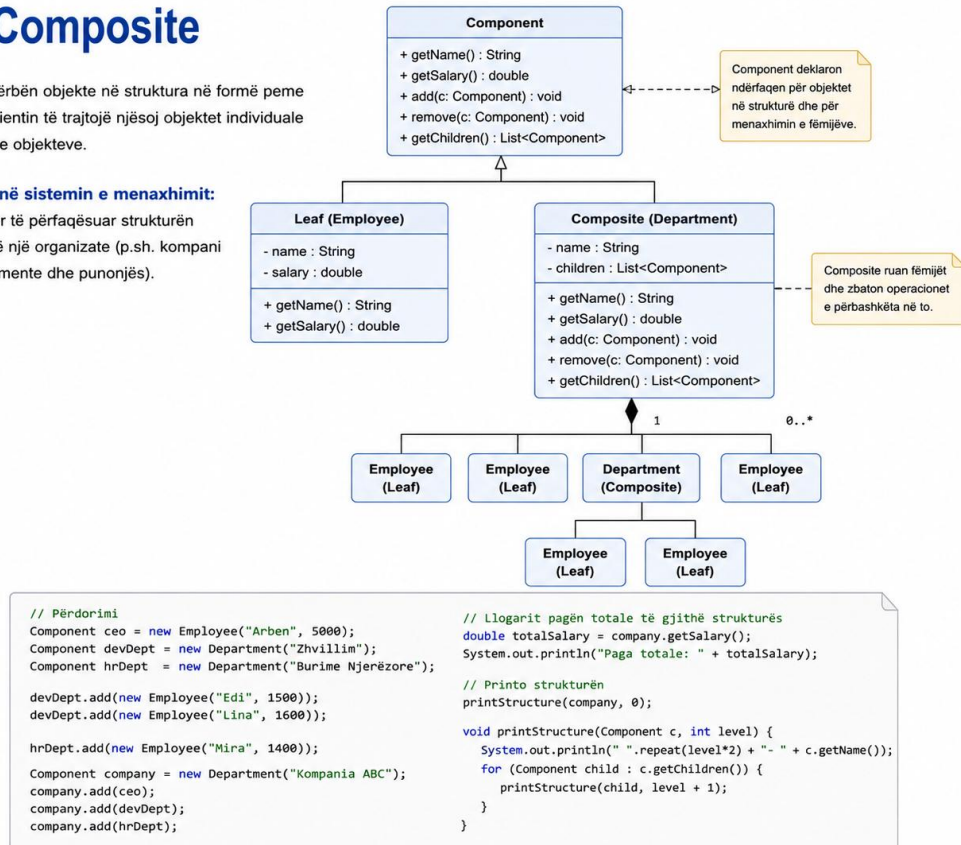


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 50: scenario9.drawio

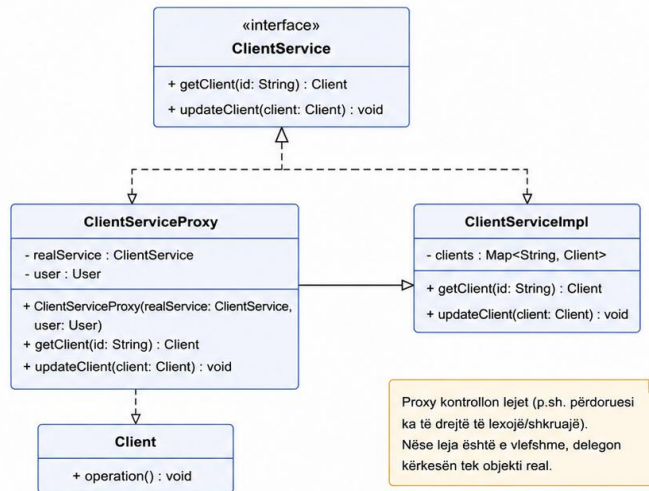
This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type.

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9 Proxy

Qëllimi: Siguron një zëvendësues ose përfaqësues për një objekt tjetër për të kontrolluar qasjen ndaj tij.

Shembull në sistemin e menaxhimit: Përdoret për të kontrolluar qasjen në të dhënat e klientit. Proxy verifikon lejet përpara se t'i delegojë kërkesën objektit real (ClientService).



```
// Klasa Proxy
public class ClientServiceProxy implements ClientService {
    private ClientService realService;
    private User user;

    public ClientServiceProxy(ClientService realService, User user) {
        this.realService = realService;
        this.user = user;
    }

    public Client getClient(String id) {
        if (!user.canRead(id)) {
            throw new SecurityException("Nuk ke leje për të lexuar klientin.");
        }
        return realService.getClient(id);
    }
}

// Metoda për përditësim
public void updateClient(Client client) {
    if (!user.canWrite(client.getId())) {
        throw new SecurityException("Nuk ke leje për të përditësuar klientin.");
    }
    realService.updattClient(client);
}

// Përdorimi
ClientService service = new ClientServiceProxy(
    new ClientServiceImpl(), currentUser);
Client client = service.getClient("C001");
```

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 51: Dfd.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or

user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

DFD LEVEL 1

Hidrosanitare Agolli – Sistemi i Menaxhimit

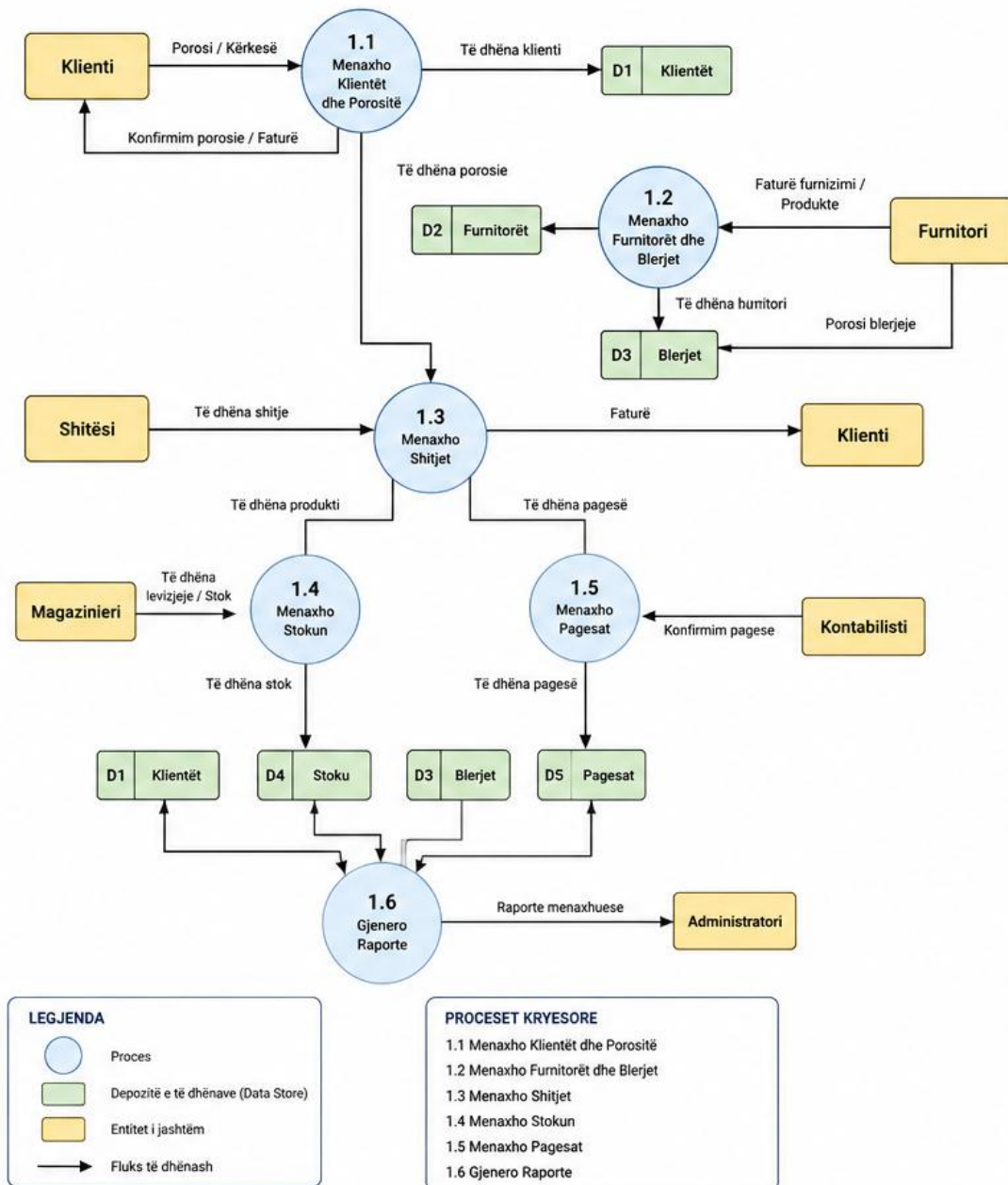


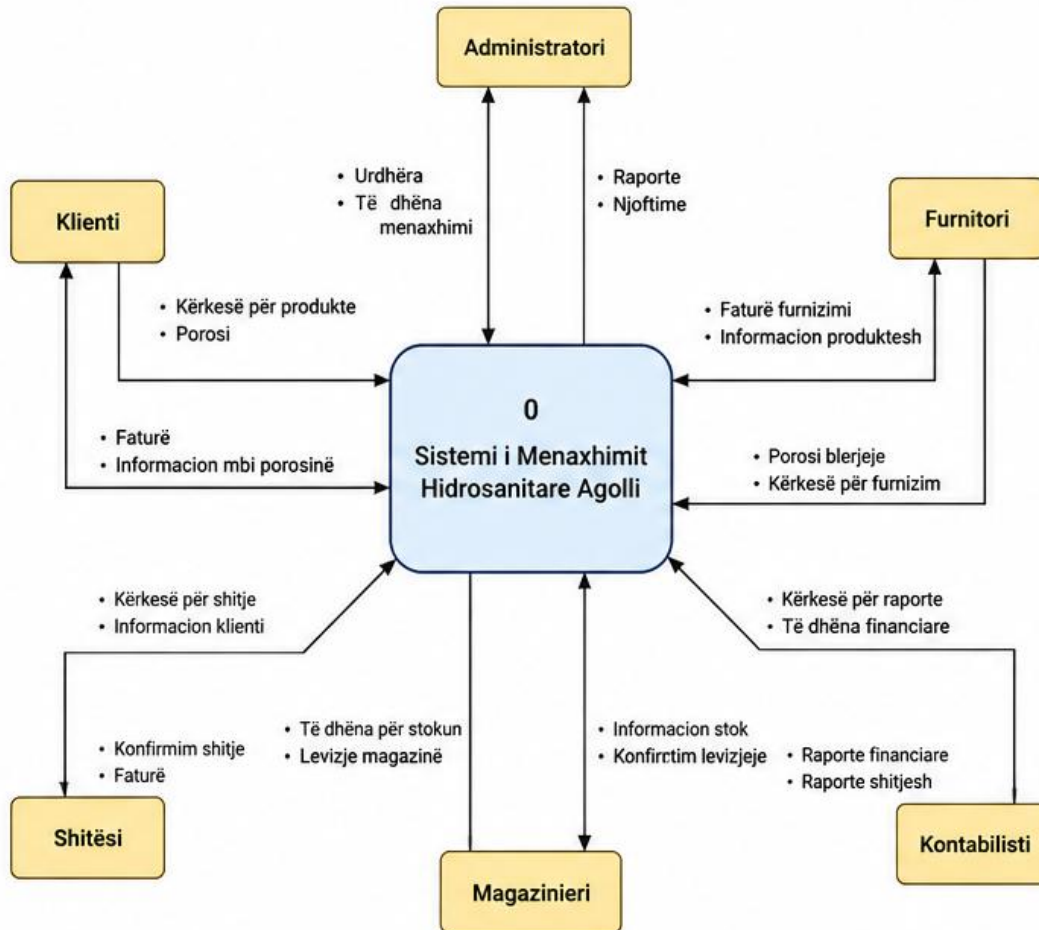
Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 52: Dfd0.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

DFD LEVEL 0 (Context Diagram)

Hidrosanitare Agolli – Sistemi i Menaxhimit



LEGJENDA



Proces



Entitet i jashtëm



Fluks të dhënash

Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 53: Packagediagram.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

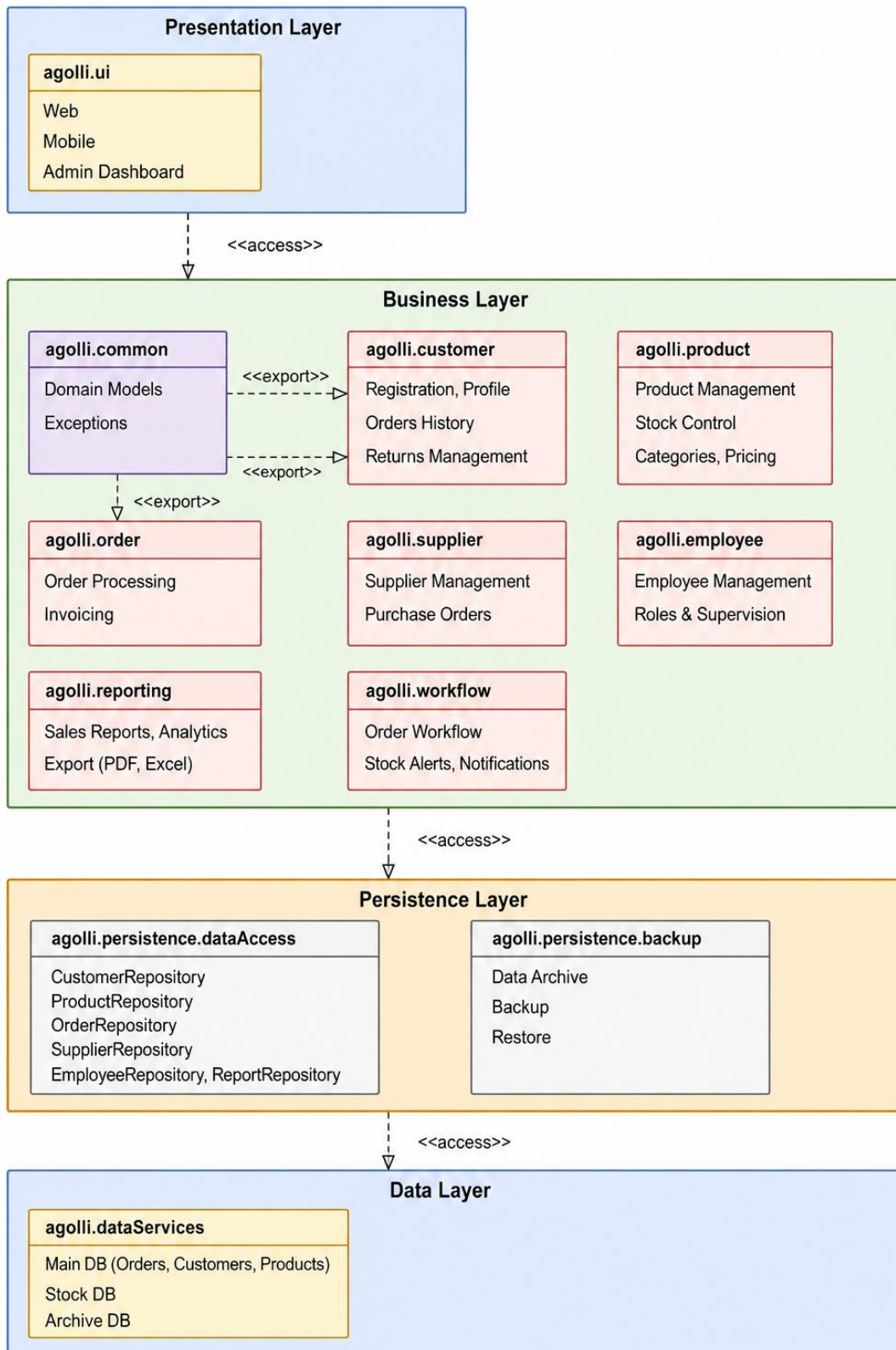


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 54: classdiagram1.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

CLASS DIAGRAM – SISTEMI I MENAXHIMIT

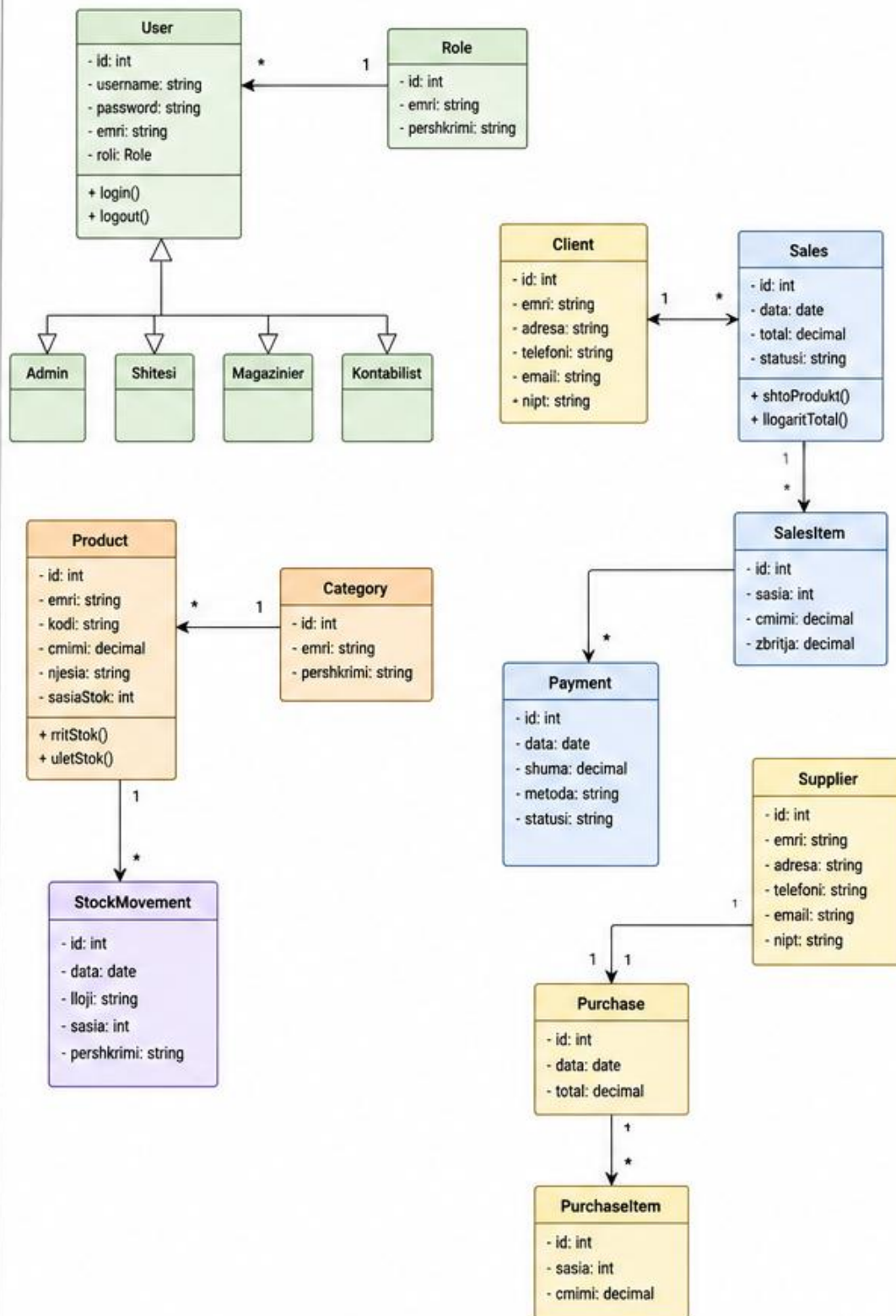


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 55: classdiagram2.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

1. Class Diagram – Menaxhimi i Stokut

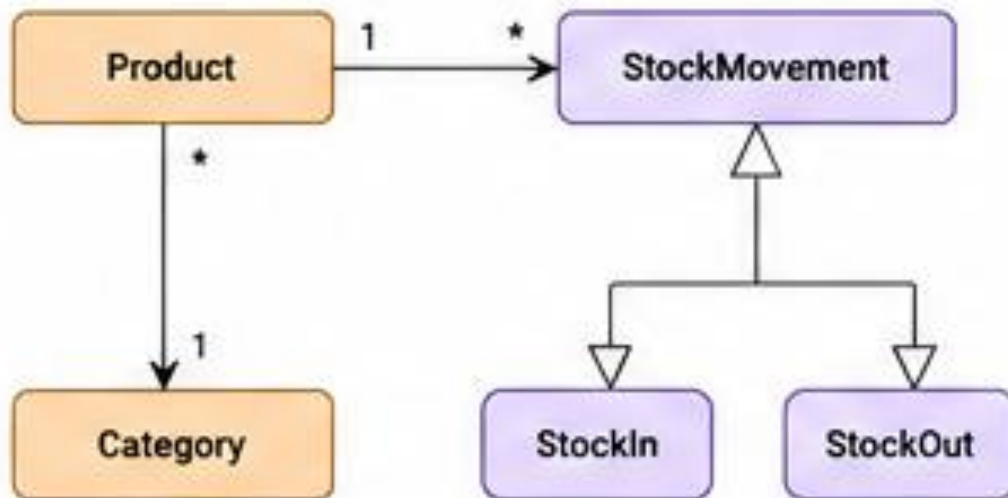


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 56: classdiagram3.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

2. Class Diagram – Menaxhimi i Klienteve

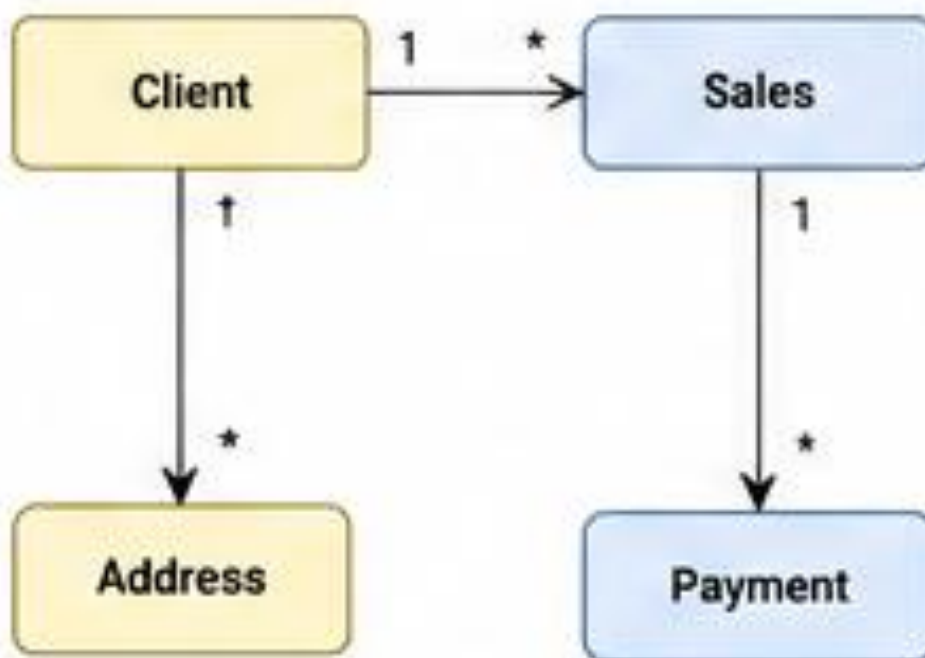


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 57: classdiagram4.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

3. Class Diagram – Menaxhimi i Furnitoreve

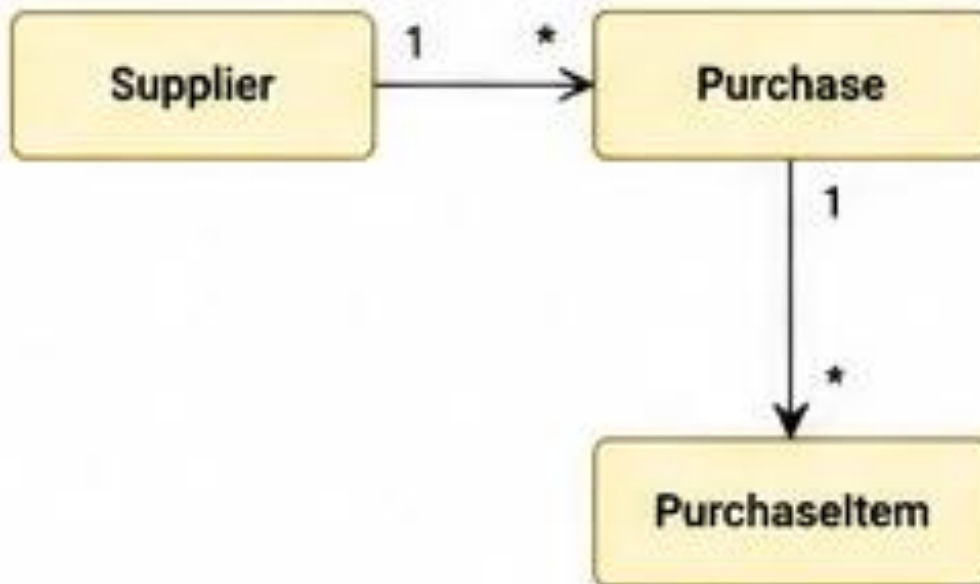


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 58: collaboration diagram

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior,

architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

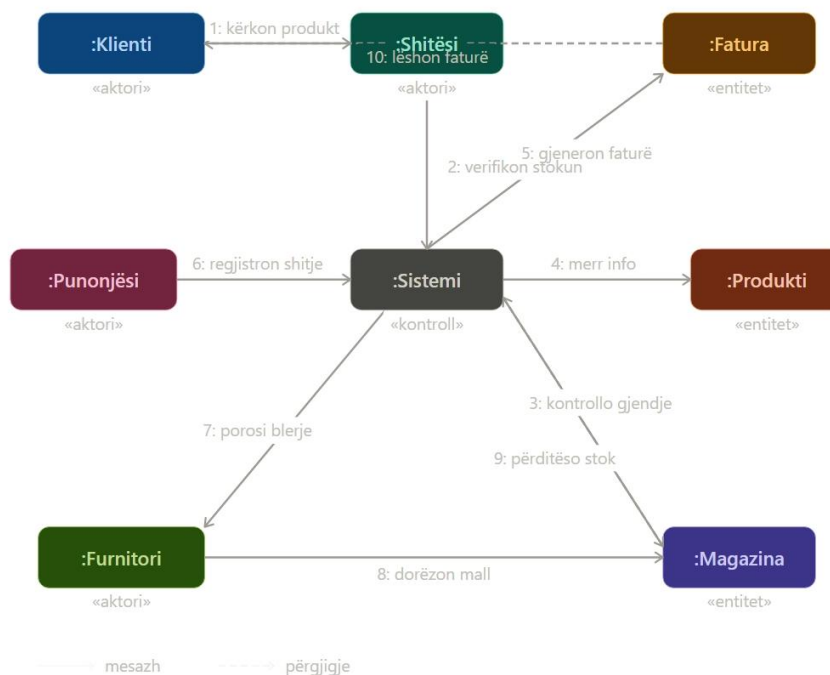


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 59: compositestructure.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type.

It contributes to understanding how the platform operates and how different software components communicate.

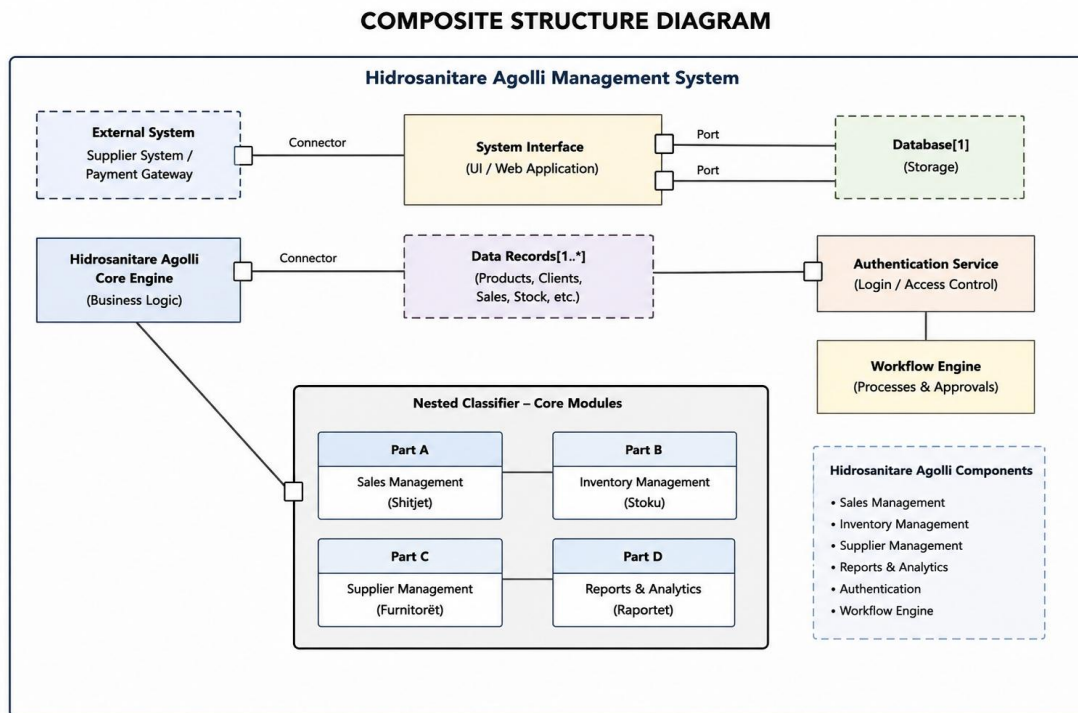


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 60: er diagram.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

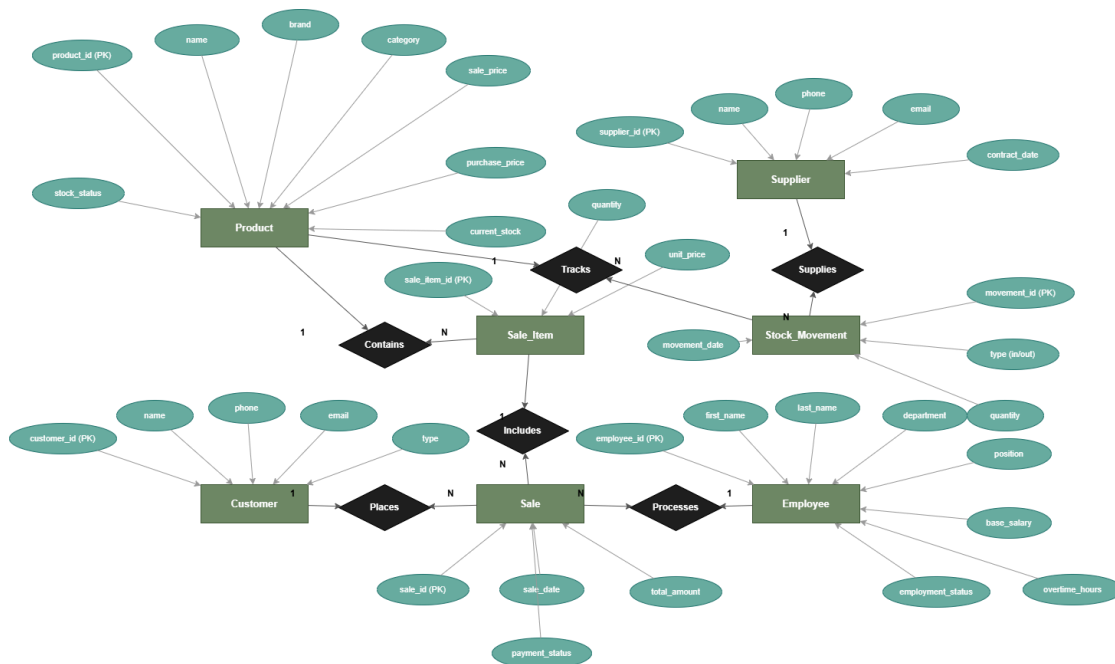


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 61: statediagram1.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

1. State Diagram – Porosia / Shitja

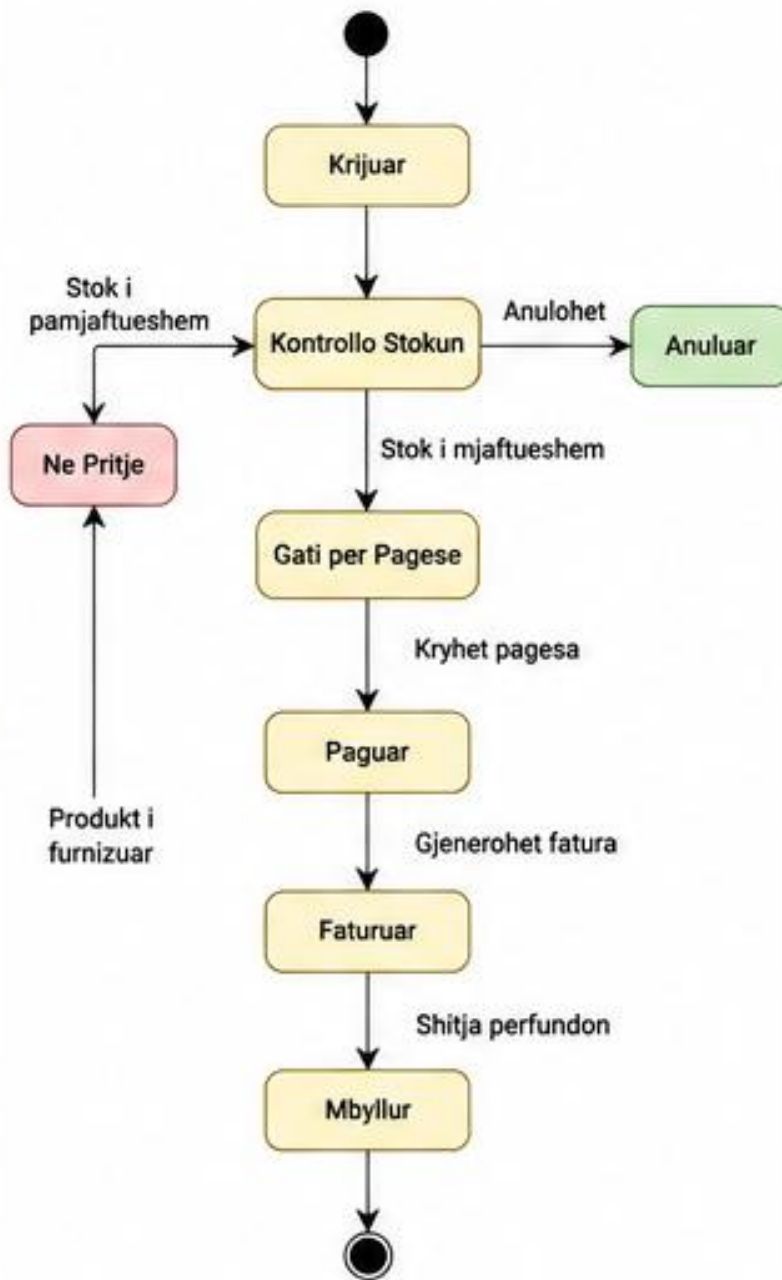


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 62: statediagram2.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

2. State Diagram – Produkti

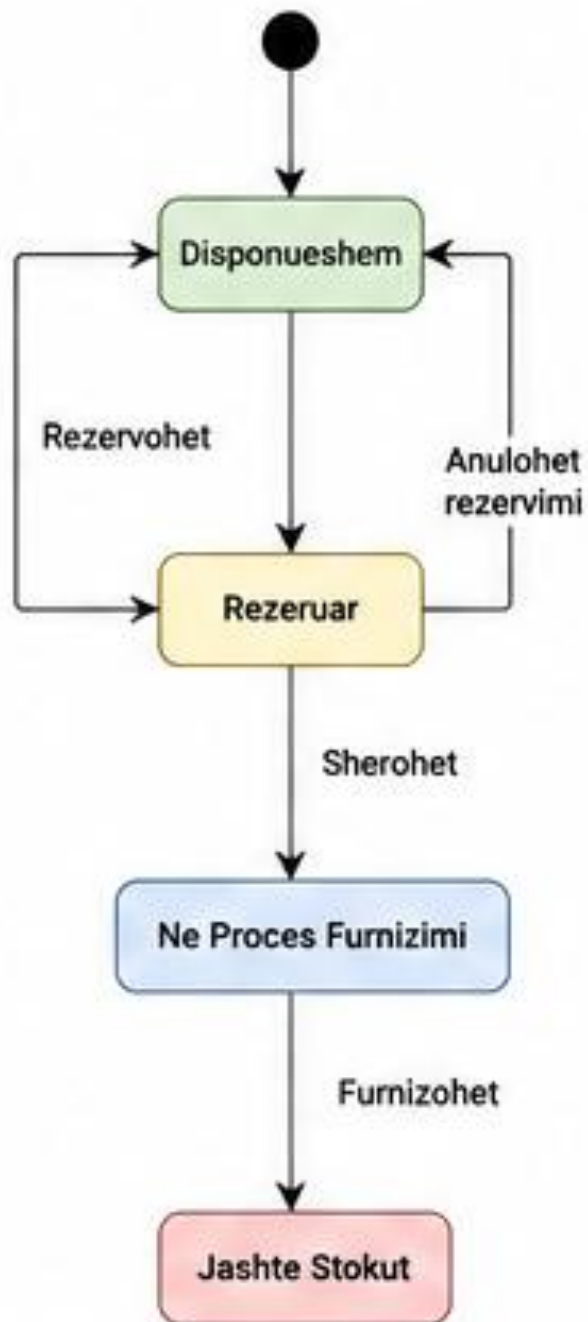


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 63: statediagram3.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

3. State Diagram – Pagesa

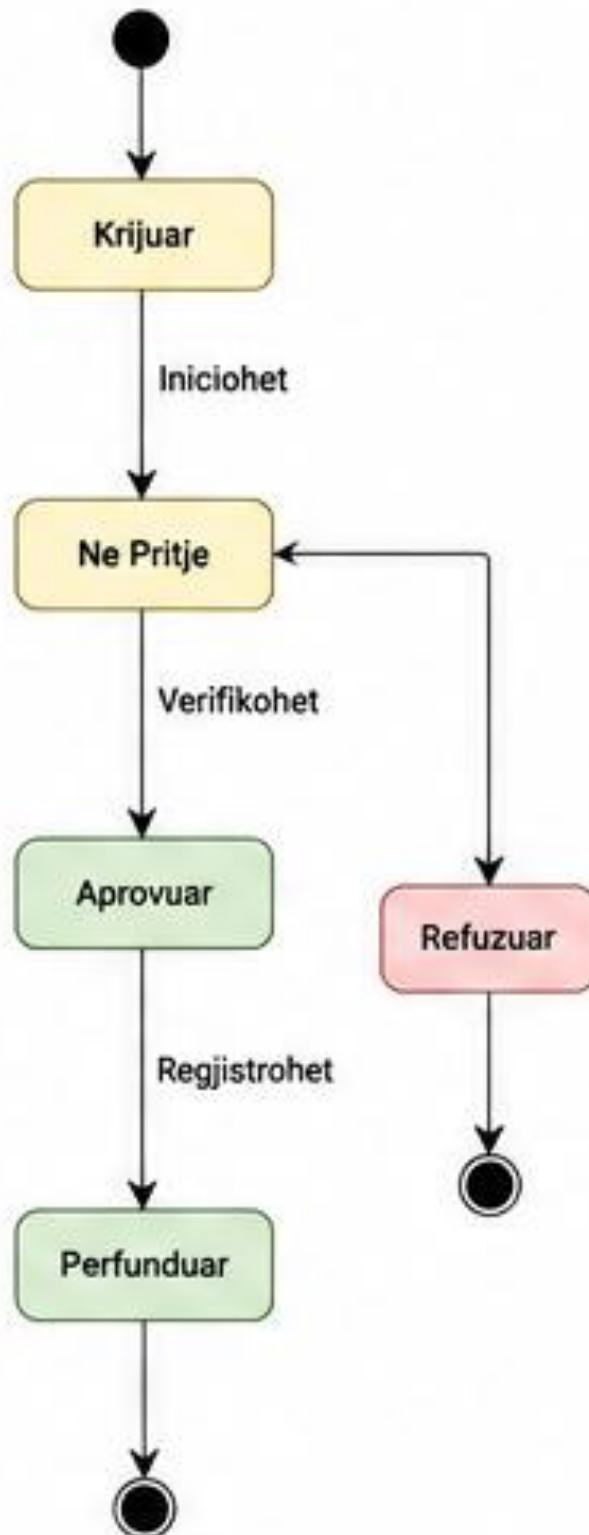


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 64: sequence1.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

SCENARIO 1: BËRJA E NJË SHITJE

Përshkrimi: Shitësi regjistron një shitje dhe sistemi përditëson stokun.

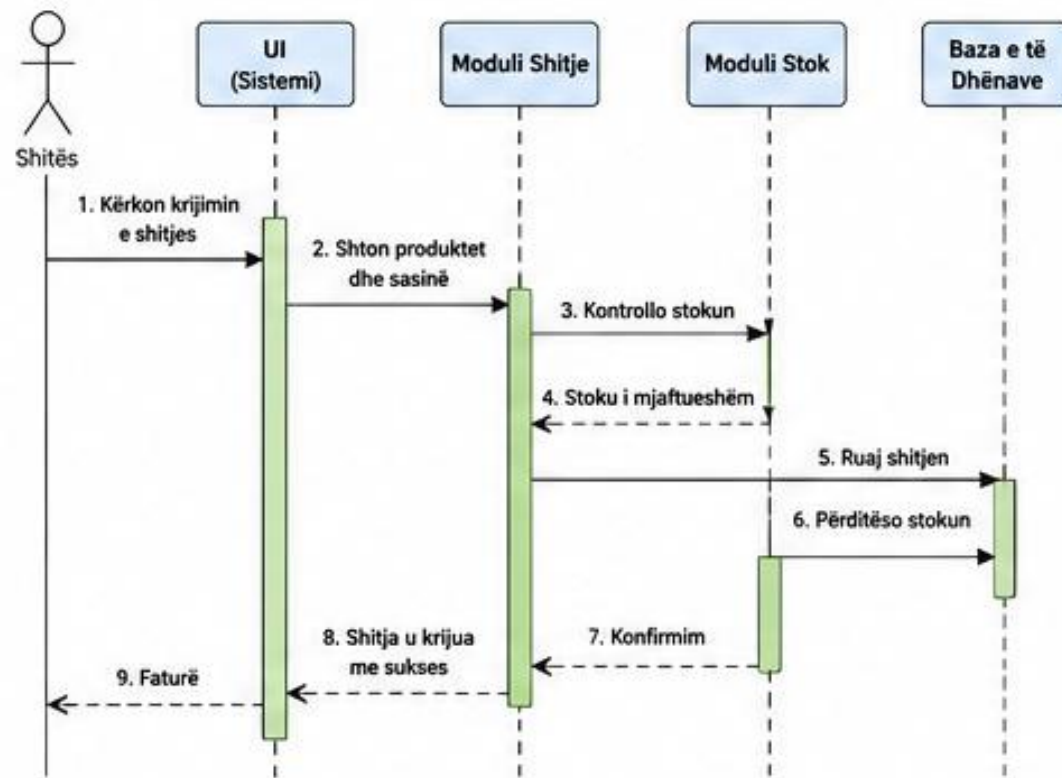


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 65: sequence2.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform

operates and how different software components communicate.

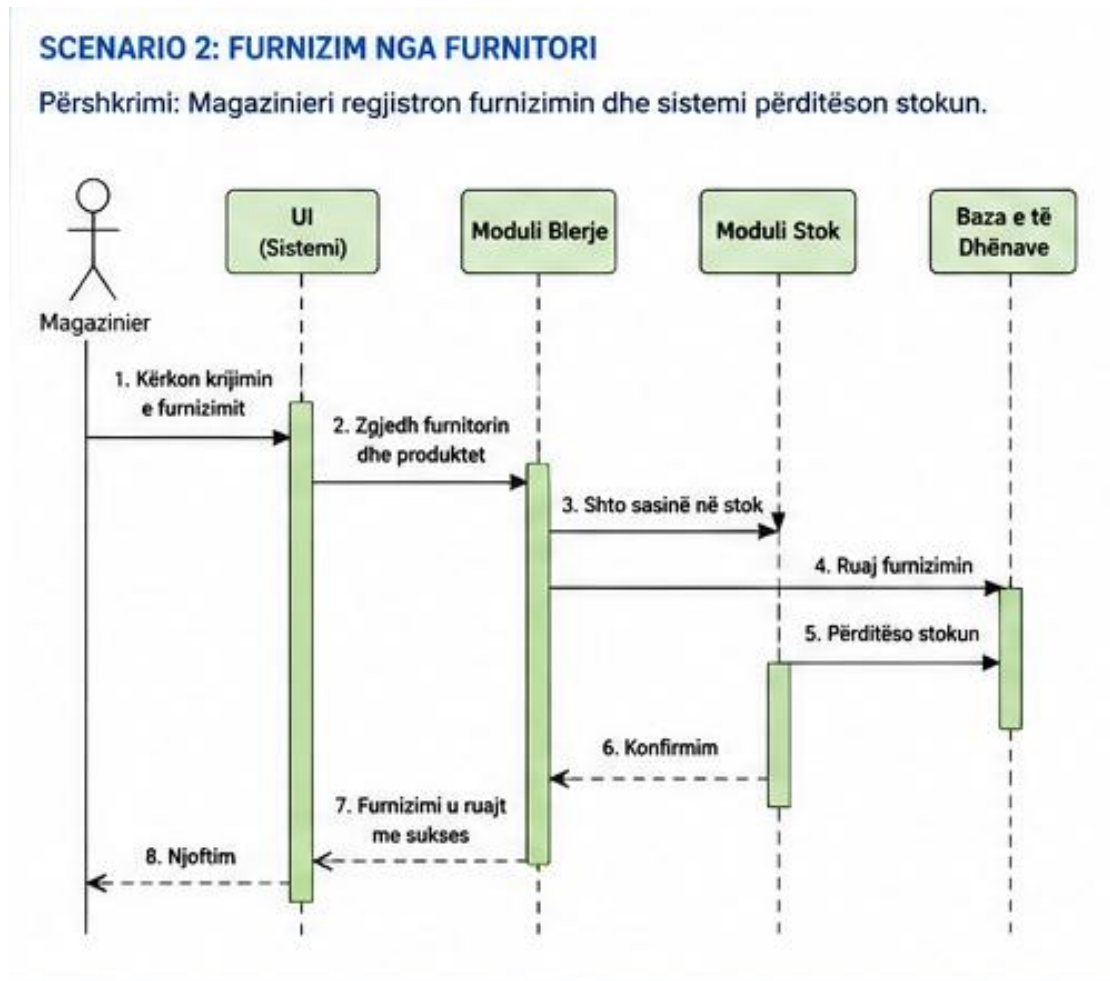


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 66: sequence3.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or

user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

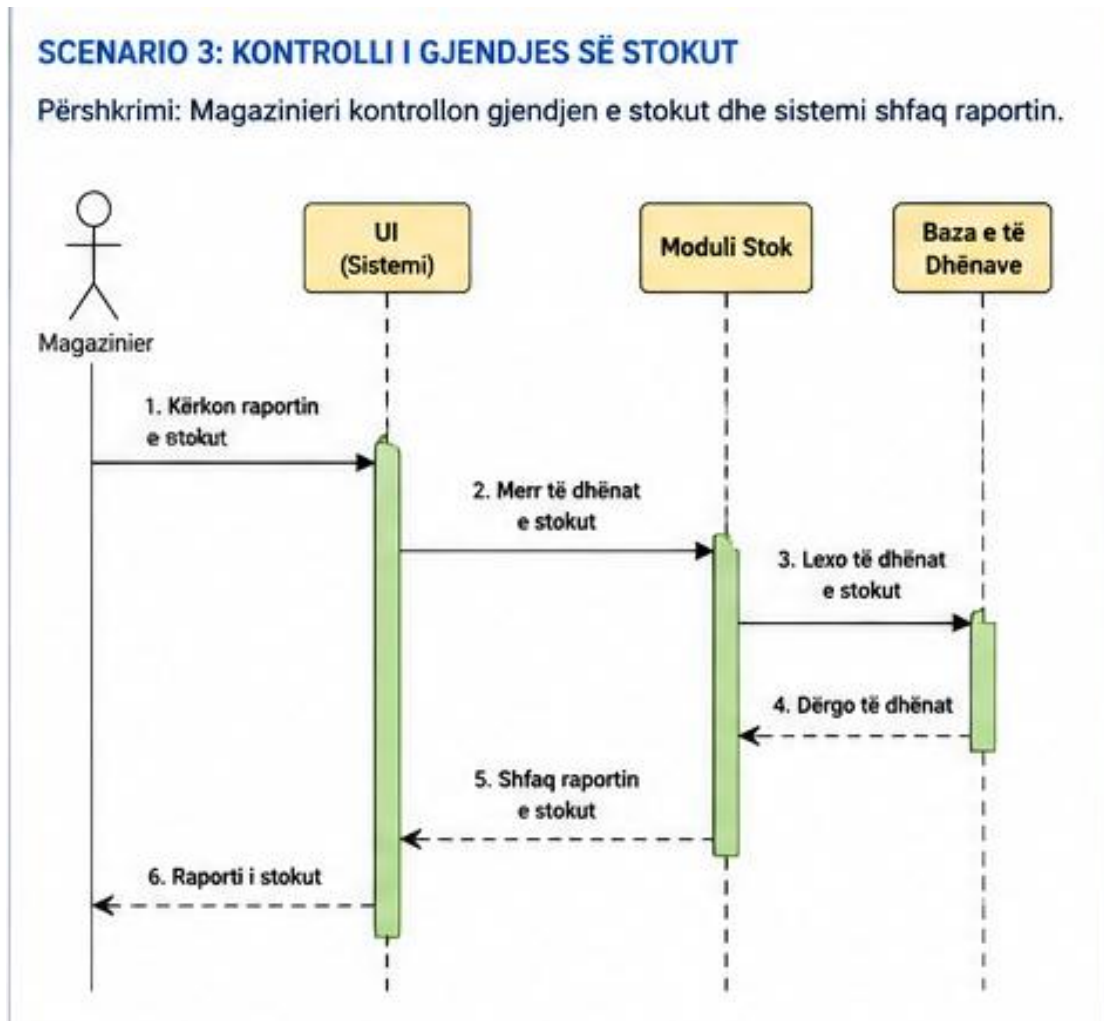


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 67: sequence4.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

SCENARIO 4: KLIENT I RI

Përshkrimi: Shitësi regjistron një klient të ri.

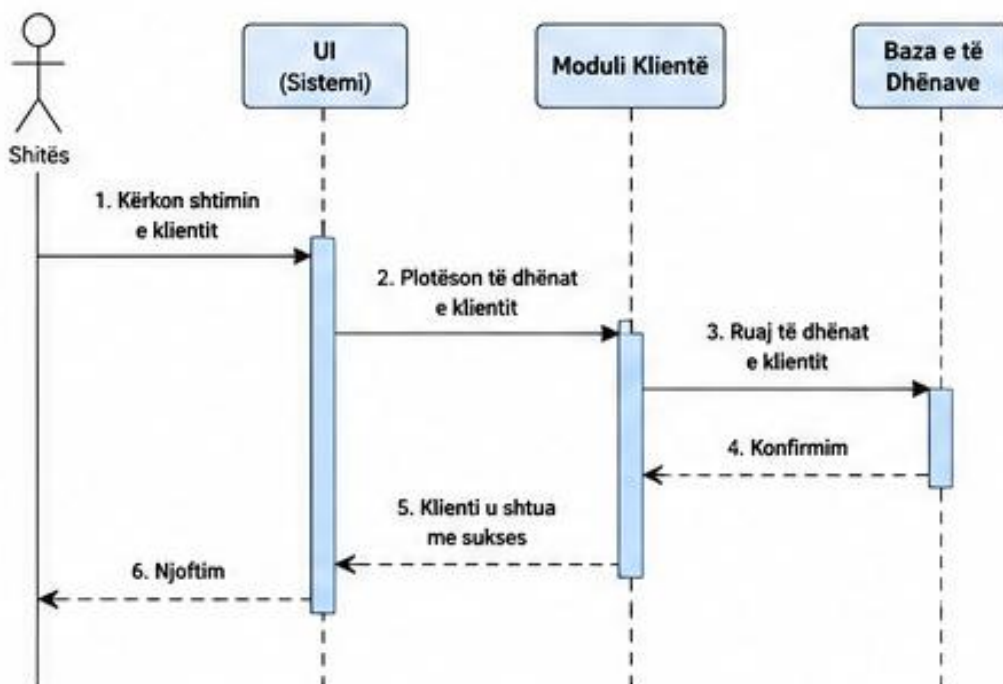


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 68: sequence5.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

SCENARIO 5: PAGESA E FATURËS

Përshkrimi: Kontabilisti regjistron pagesën e një fature nga klienti.

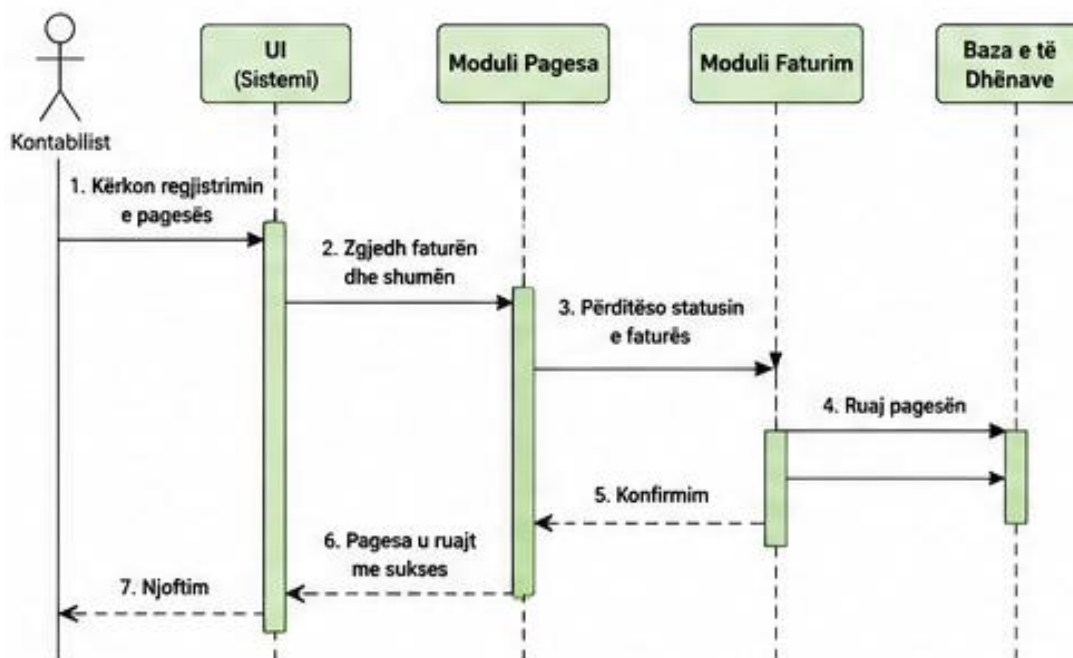


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 69: sequence6.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

SCENARIO 6: GJENERIMI I RAPORTEVE

Përshkrimi: Administratori gjeneron raportin mujor të shitjeve.

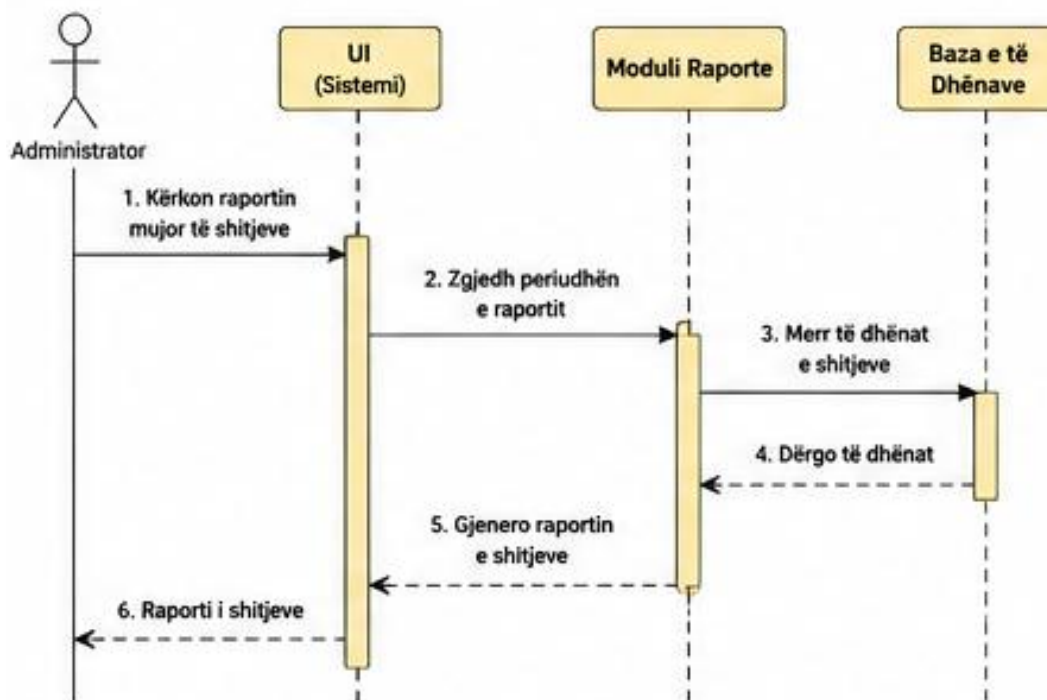


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 70: uml.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

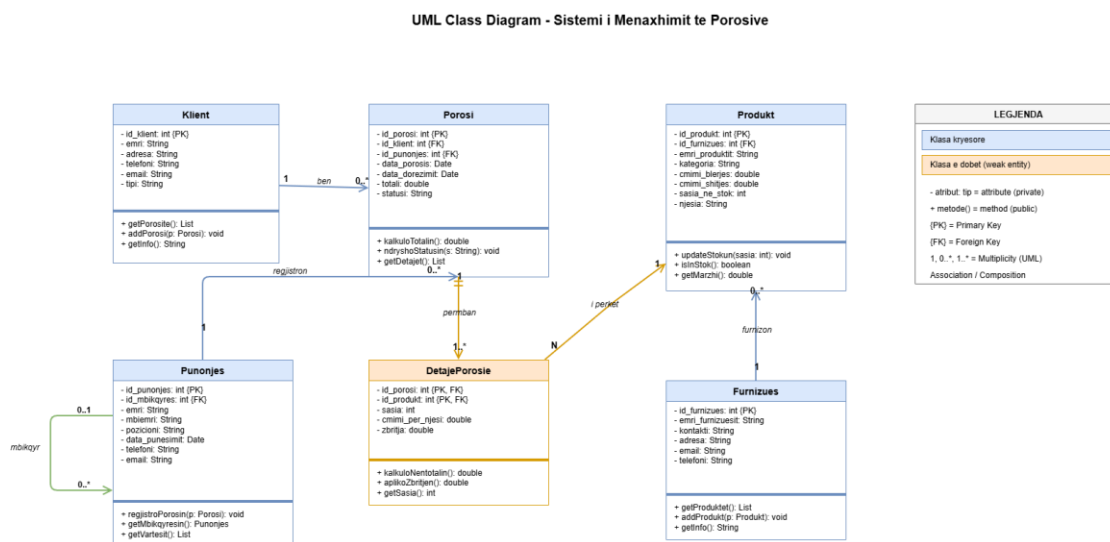


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

Diagram 71: Usecase.drawio

This diagram represents part of the system analysis and design process for the Hidrosanitare Agolli Management System. The figure illustrates system behavior, architectural structure, business logic, workflows, or user interaction depending on the specific diagram type. It contributes to understanding how the platform operates and how different software components communicate.

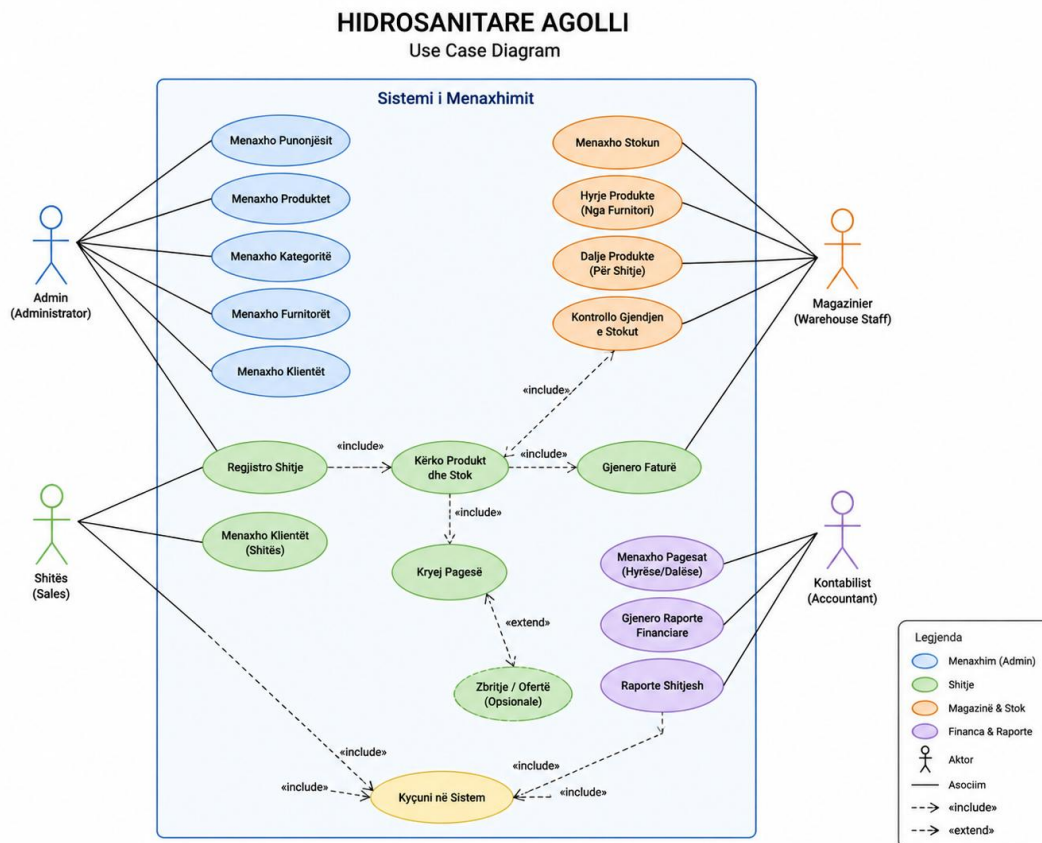


Figure Description: The diagram above was extracted directly from the official repository and included as part of the project documentation.

6. User Interface Prototype

The repository also contains several user interface prototype images that demonstrate the expected layout and user experience of the system. These interfaces are intended to simplify business management tasks and improve usability for administrators and employees.

7. Database and Data Flow Design

The repository includes multiple diagrams related to database architecture, entity relationships, and data flow modeling. These diagrams help define how data is stored, processed, validated, and transferred between modules inside the system.

8. Conclusion

The Hidrosanitare Agolli repository demonstrates a complete software analysis and system design project that includes a broad range of software engineering artifacts. The diagrams, prototypes, and documentation included in the repository provide a detailed understanding of the platform architecture, business logic, workflows, and system requirements.